Persistent Specialization and Growth: The Italian Land Reform*

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Abstract

The impact of land redistribution on structural transformation is ambiguous. While large landowners may hinder industrialization by restricting access to education, larger farm scale can facilitate mechanization and productivity growth. This study uses novel fine-grained data to examine the long-term effects of the 1950 Italian land reform, which redistributed land from large landowners to landless farmers. Employing two difference-in-differences strategies, we find that the reform significantly slowed industrialization in affected municipalities, which, fifty years after the reform, exhibited an agricultural employment share approximately 70% higher than the estimated counterfactual scenario. Reductions in agglomeration and occupational mobility emerge as key mechanisms, while education seemingly played a limited role. Finally, we show that the reform significantly hindered the overall economic growth of affected municipalities between 1970 and 2000.

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1 Introduction

The distribution of landownership has historically played a central role in shaping economic development trajectories and land reform policies have been implemented in various contexts with the dual objective of improving economic efficiency and reducing inequality. However, there is a well-recognized theoretical tension in how landownership concentration affects economic development. On the one hand, it is often associated with limited access to education and slower structural transformation, as entrenched elites may have little incentive to invest in broad-based human capital. On the other hand, large landowners can exploit economies of scale and more easily access the capital needed for mechanization, thereby increasing agricultural productivity; in rural labor markets where they exercise monopsony power, they may reduce agricultural employment and push workers toward other sectors.

As a result, whether such interventions can produce a more efficient agricultural sector, structural transformation, and local economic growth remains an open empirical question. While considerable attention has been paid to the implications of landownership inequality within agriculture, its broader and long-term effects on structural change and economic development remain less well understood. This paper examines how a large-scale land reform in postwar Italy influenced sectoral employment patterns in the short run and contributed to long-term structural transformation and economic growth.

In the 1950s, the Italian government implemented a land reform aimed at redistributing wealth, increasing agricultural efficiency, and securing political consensus. Our empirical analysis relies on newly digitized historical records from several sources to produce a novel and unique dataset. The digitized sources include a 1948 census of agricultural land, the 1936 population census, official expropriation acts, and the earliest available income estimates for Italian municipalities. These were combined with several decennial population and agriculture censuses, historical maps, and more recent income estimates to produce a rich, fine-grained dataset tracking the evolution of a variety of socioeconomic outcomes over 65 years. We first show that affected areas experienced a reduction in the Gini index of land ownership and in the share of large plots. To identify the causal effects of land redistribution on structural transformation, we estimate a difference-in-differences model, tracking the share of individuals employed in agriculture and manufacturing across several census waves. We build two main samples: the first is based on administrative boundaries, while the second relies on propensity score matching to produce balanced treatment and control groups. These models account for time-invariant municipal characteristics and shared time trends, and the absence of pre-treatment differential trends supports the validity of our approach.

We find that expropriation and redistribution led to a substantial and persistent increase in agricultural employment, accompanied by a corresponding decline in manufacturing activity. Our estimates for the impact on the share of agricultural employment range between 2.9 and 4 percentage points, depending on the preferred sample, and are statistically significant. Fifty years after the reform, treated areas exhibited an agricultural employment share of 14%, approximately 70% higher than our estimated counterfactual scenario. The share of employment in manufacturing decreased by a very similar amount, while effects on employment in the tertiary sector are smaller and not statistically significant in most specifications. These findings are robust to multiple sensitivity and robustness checks and generally consistent across different geographical regions, with the Center-North sample showing a greater negative impact on manufacturing.

What explains these effects on occupational composition? Contrary to the view that reductions in land inequality fosters human capital accumulation (Galor et al., 2009; Cinnirella and Hornung, 2016), we find no overall significant effects on educational attainment (i.e., literacy rate and high school completion). This null result is itself informative, as it suggests that land reform did not catalyze long-term investments in education, and that other factors, such as the expansion of the public education system, may have played a more decisive role in shaping human capital outcomes. Instead, we show that treated areas became less densely populated and less concentrated around towns and cities, suggesting that weakened agglomeration forces may have slowed industrialization (Breinlich et al., 2014; Martin and Ottaviano, 2001). Moreover, we argue that a reduction in occupational mobility reinforced the long-term effects of the reform on sectoral composition. Historical sources, prior literature on intergenerational occupational mobility (Long and Ferrie, 2013), and our own analysis of Italian survey data provide suggestive evidence of a relationship between landownership and occupational mobility, with landowning farmers being much more likely to transmit their occupation to their children. We also explore the reform's impact on migration and fertility patterns, and on agricultural productivity. While data availability is limited, we do not find strong evidence supporting the hypothesis that these channels played a significant role in mediating the reform's impact.

Finally, given the ambiguous relationship between agricultural specialization and income levels, we assess whether the reform fostered or hindered local economic prosperity. We digitize the earliest available income estimates for Italian municipalities and produce new municipal-level measures of growth between 1970 and 2000. We estimate the causal impact of the reform using a propensity score matching method based on pre-reform land inequality, agricultural productivity, and other sociodemographic characteristics to produce balanced treatment and control groups. Our findings indicate that areas affected by land redistribution experienced significantly lower long-run growth, approximately 20 percentage points lower between 1970 and 2000, compared to a baseline increase in per capita income of about 100%.

Our first contribution relates to the empirical literature on the relationship between the agricultural sector and structural change and economic growth. Matsuyama (1992) and Gollin (2023) underscore the foundational role of agriculture in structural transformation, arguing that shocks originating in the agricultural sector, like improvements in agricultural productivity, can have profound and lasting effects on the reallocation of labor and lead to economic development. Existing studies, such as Bustos et al. (2016, 2020); Asher et al. (2022) and Gollin et al. (2021), focus on productivity-enhancing shocks (e.g., better crops or irrigation improvements) that induce labor reallocation out of agriculture and accelerate structural transformation, particularly when they reduce labor demand and encourage capital deepening and population agglomeration. However, shocks to agriculture can also slow down structural transformation: we show that the Italian land reform represented a labor-absorbing shock and led to a persistent increase in agricultural employment and a deceleration of structural transformation. Rather than freeing up labor, land redistribution expanded smallholder farming without inducing technological upgrading, population agglomeration, or capital intensification, resulting in a long-term retention of labor in agriculture. We also don't find consistent evidence of impacts on human capital accumulation, which is considered an enabling factor for structural transformation.[1] This is consistent with findings from a major land reform in Peru (Albertus et al., 2020), which similarly failed to improve educational outcomes. This lack of human capital accumulation likely contributed to the persistence of agricultural employment and the deceleration of structural transformation, supporting the idea of education as a key constraint on the reallocation process.

Our work also contributes to the strand of literature examining the role of landownership in shaping long-run economic development. A substantial body of research has shown that the distribution of land affects labor allocation, investment decisions, and the pace of structural transformation (e.g., Besley and Ghatak, 2010; Adamopoulos and Restuccia, 2020). Within this literature, a central tension emerges. On the one hand, concentrated landownership has been linked to lower investments in education and delayed structural change, as powerful landowning elites may have limited incentives to support broad-based development (Galor et al., 2009; Cinnirella and Hornung, 2016). On the other hand, large landowners may raise agricultural productivity through mechanization and economies of scale (Foster and Rosenzweig, 2022) or limit agricultural employment by exerting monopsony power in rural labor markets (Martinelli, 2014). Despite a rich theoretical foundation, the long-run empirical consequences of land redistribution policies for structural transformation and local economic growth remain relatively underexplored. We contribute to this literature by providing empirical evidence that land redistribution led to a persistent increase in agricultural employment and a deceleration of structural transformation, without measurable improvements in educational attainment. In the long run, treated areas experienced significantly lower income growth, suggesting that redistributive land policies can result in unintended and lasting distortions in local economic development.

[1] Porzio et al. (2022) show that higher levels of education facilitate the reallocation of labor toward more productive sectors and contribute to long-run growth.

We also add to the literature on land reforms. Historically, land reforms have taken a variety of forms, including redistribution (Albertus et al., 2020, Adamopoulos and Restuccia, 2020, Albertus, 2025), land titling formalization (De Janvry et al., 2015), land granting (Mattheis and Raz, 2021), and changes in the organization of production (Montero, 2022). The quality of implementation has also been shown to vary significantly across contexts and to be important in shaping long-run outcomes (Lipton, 2009; Besley et al., 2016; Galán, 2024). According to historical accounts, the Italian land reform focused primarily on redistribution and was carefully implemented, making its long-term effects particularly informative. Consistent with these accounts, we find that areas with low agricultural productivity and high land inequality were more likely to experience intensive expropriation, and that the reform successfully and persistently reduced land concentration. We generate new evidence on the short-, medium-, and long-term impacts from a large-scale, well-implemented case of land redistribution, linking fine-grained historical information on expropriations to long-run sectoral and income outcomes. Our findings of persistently increased agricultural employment together with reduced economic growth highlight how even carefully designed reforms in market economies[2] can produce enduring frictions in labor reallocation and local development.

Finally, we contribute to the growing literature on land and agricultural policies in Italian history. Carillo (2021) analyzes Mussolini's Battle of Grain and finds that the policy favored industrialization thanks to its effects on education returns. Martinelli (2014) studies the historical latifundia system and shows that high land concentration in interwar Italy was associated with lower agricultural efficiency, driven in part by the monopsonistic power of large landowners over rural labor markets. In more recent work, Martinelli and Pellegrino

[2] In the Philippines, where a ceiling on landholdings and a prohibition on transfers were imposed, Adamopoulos and Restuccia (2020) documents reductions in average farm size and agricultural productivity, arguing that land transfers could have dramatically mitigated these negative impacts. In our context, a ban on land transfers was imposed but soon lifted, and no ceiling on farm size was ever implemented (see Section 2). Although 7 percent of the original land recipients no longer owned land by 1960 (King, 1973), and 17 percent by 1974 (Angeli and INSOR, 1979), we observe persistent reductions in the share of large farms, higher agricultural employment, and negative long-term effects on income in reformed areas. This suggests that even when land transfers are permitted, inefficiently small farms are likely to persist, and markets may fail to generate optimal reallocation of resources.

(2024) documents a negative relationship between land inequality and structural transformation during the country's post-war growth, while highlighting important non-linearities related to farm size. Three recent papers have studied the 1950 land reform. Caprettini et al. (2024) examine the political consequences of land redistribution, providing evidence that redistribution shaped long-run political preferences. The other two contributions focus on the reform's impact on economic development. Albertus (2023) focuses on the regions of Tuscany and Lazio and offers evidence of negative long-term effects on well-being measured in 2011, conducting a geographic discontinuity analysis. Percoco (2018), instead, finds opposite results for three regions in the South using a Kitagawa-Oaxaca-Blinder approach. This paper builds on both studies and addresses some of their limitations.[3] Our empirical approach and data availability allow us to better address concerns of systematic differences in the outcome levels and spatial contiguity of treated and untreated municipalities. Additionally, we provide a much more comprehensive evaluation of the reform, providing evidence for all eleven regions with available expropriation data.[4] The additional data and robust approach produce novel findings and a nuanced picture: we find that the increase in agricultural specialization is consistent in the Center-North and South, but observe slightly different patterns for industrialization, where the South appears to lose tertiary employment instead of manufacturing. We also show that impacts grow over time, suggesting that the initial shock put treated areas on different development paths. Finally, our analysis of income growth shows that the reform had negative long-term economic impacts beyond specialization.

The rest of the paper is organized as follows. Section 2 reviews the historical background. Section 3 illustrates our data, and Section 4 presents our identification strategy. Section 5 describes the main results, discusses their robustness, and analyzes the potential mechanisms. Section 6 investigates the relationships between land reform and economic growth in the long run. Finally, Section 7 contains concluding remarks.

[3] For a more detailed discussion, see Appendix Section F.

[4] The only region that experienced land redistribution and is not included for lack of data is Sicily. More details in Section 2.

2 Historical Framework

Background

Following World War II, rural agricultural workers in Italy, particularly in the South, faced severe economic hardship, with working conditions resembling a quasi-feudal system. Regional disparities were stark: while Calabria's per capita income was approximately half the national average, Piedmont's reached 174% (Ginsborg, 2003). Government opposition highlighted the underdevelopment of areas like Calabria: a 1949 Communist Party regional assembly stated that 90% of municipalities lacked proper school buildings, 85% had no drainage systems, and 81% lacked aqueducts. Moreover, nearly half the population was illiterate (Ginsborg, 2003).

By the late 1940s, worsening economic conditions and systemic inequalities led to widespread land occupations by agricultural workers. These movements were fueled by deep-seated grievances against absentee landowners, whose inaction and exploitative labor arrangements had persisted for decades.[5] These conditions triggered violent clashes with the police as the government sought to contain the unrest (Ginsborg, 2003).

Crafting the law

To avoid further escalation of social unrest, the Christian Democrats (i.e., the ruling party since the 1948 elections) decided on a redistributive plan and, in the first semester of 1950, presented a land reform to the Parliament (N. 977). Reformed areas were identified with the help of agrarian technicians. A modified proposal was enacted in October: law n.841, called *Legge Stralcio*[6] (Bagnulo, 1976). The expropriation rule (see Appendix Figure A1) used measures of inequality and productivity to assess the amount of land to be taken from each landowner, aligned with the reform's stated objectives of promoting redistribution and fos-

[5] Martinelli (2014) suggests that large landowners enjoyed sizable market power over labor workers. King (1973) notes that "The southern landlord generally contributed nothing more than the land to the contract, and the peasant had to pay rents in cash or kind that ranged from 25% to 60% of the value of the crop. [...] The majority of latifondisti were absentee landlords, [...] only visiting their estates for hunting purposes."

[6] *Legge Stralcio* translated to "excerpt of law", alluded to the fact that more would be done to address the social and distributional issues of the affected areas.

tering economic development. As a result, the expropriation mechanism was designed to be both selective and automatic, minimizing discretionary interpretation by local reform bodies and ensuring uniform treatment across regions. By applying standardized tax-based thresholds, the system imposed a heavier expropriation burden on extensive, low-productivity estates while sparing more intensive, high-yield properties-thus aligning the reform with both equity and efficiency objectives. An implicit objective of the reform was to contrast the rhetoric of the Italian Communist Party, which fomented and led many of the revolts and land occupations (Ginsborg, 2003).

On the Ground: The Implementation

The reform that was enacted had a broad scope, potentially impacting 8.5 million hectares, approximately one-third of the country's total land area, as shown in Appendix Figure A2.

Ultimately, nearly 700,000 hectares were redistributed to approximately 120,000 families. The implementation of the reform at the local level was assigned to the *Enti di Riforma* (reform bodies), with one authority assigned to each designated reform area. These institutions were responsible for managing applications and overseeing the reform process.

On the redistribution side, the assignment of land to beneficiaries was regulated by broad national principles, but operationalized differently by each *Ente di Riforma*. Laws required that beneficiaries be heads of rural households with no land or insufficient holdings to sustain family labor. However, beyond this, each reform body established its own priority criteria. For instance, the Puglia-Basilicata-Molise authority ranked landless laborers and agricultural wage workers highest, while smallholders were ranked lower. In Tuscany, priority was often given to sharecroppers already living and working on the targeted land (see Marciani, 1966). Across regions, low-income applicants and residents of the same municipality were systematically favored, minimizing forced relocations and aligning reform goals with local labor markets.[7] Although there was a formal application procedure, access was limited

[7] According to Pezzino (1977) Calabria was the region with the highest share of land reform beneficiaries coming from municipalities different from those where the assigned land was located. As of September 30, 1960, 17.3% of assignees came from neighboring municipalities, 3.5% from non-adjacent municipalities within the same province, and only 0.4% from municipalities in other provinces. Even in this case, the data suggest that

to a well-defined population segment, and the allocation was structured through regionally specific rankings rather than open competition. Rural workers who were allocated a plot could purchase it through a loan repayable over 30 years. Initially, early repayment and land rental or sales were not permitted, but these restrictions were formally lifted in 1967 (King, 1973). The redistributed plots varied in size: smaller plots, known as *quota*, were intended to supplement household income, while larger plots, called *podere*, were meant to support independent farming operations. However, the land allocation procedures differed across regions, resulting in inconsistencies in implementation.[8]

Expropriated landowners also faced restrictions to prevent a swift return to the pre-reform status quo: they were prohibited from purchasing land for six years. Additionally, the implementation of the reform included measures to prevent landowners from evading expropriation or exploiting the process for personal gain. Two key provisions ensured this. First, land distribution decisions were based on ownership records from 1949, preventing landowners from artificially dividing their estates among family members or engaging in fraudulent land transfers. Second, compensation was determined using 1947 tax returns (Bandini, 1952), ensuring that payments reflected historical valuations rather than inflated market prices. Expropriated landowners were compensated with 25-year fixed-rate government bonds at approximately one-third of the market value, according to Marciani (1966).

To address concerns that the reduction in operational scale due to expropriations might hinder investment capacity and productivity, assignees were required to join cooperatives for twenty years from the signing of the sales contract (Bandini, 1952). Ideally, these cooperatives would undertake high-cost investments in equipment and infrastructure to support the processing and commercialization of agricultural products. However, in practice, this requirement served more as a means of education and training than as an essential economic measure (Angeli and INSOR, 1979). One cooperative, Coldiretti, would later play an im-

proximity was generally favored in the allocation process.

[8]Several scholars, like Ginsborg (2003) and King (1973), remarked that political ideology played a role and that applicants with known communist sympathies were penalized. This is further confirmed by Pezzino (1977), who describe the allocation process in Calabria and emphasize the influence of political ideology (p.101). Caprettini et al. (2024) argues that this was less prevalent in the Center and North of Italy.

portant role in shaping agricultural policy and dynamics at the national level. However, its reach and the services it provided were not exclusive to reformed areas or to landowners.

A Short-Term Assessment

The implementation of the Italian land reform of the 1950s attracted considerable attention. Prinzi (1956) and Rossi-Doria (1958) described the implementation process and its short-term effects. King (1973) and Angeli and INSOR (1979) provide an overall assessment of the success of the reform in attaining its main goals 20 years after the law was signed. According to Angeli and INSOR (1979), 83% of the original 120,000 firms were operating in 1974, and 80% of them were still cultivated by the original assignees or their descendants.[9] Rural workers grew their possessions: an additional 170,000 hectares of land not affected by the reform were cultivated in 1974.

Unlike the mechanical effects on land distribution, the effects on productivity are ambiguous due to the countervailing effects of land improvement, agency realignment, and scale reduction. The available evidence suggests that the reform significantly improved productivity in the affected areas. King (1973) shows that productivity growth in the reformed areas was higher than the national average between 1953 and 1963 (see Appendix Table A1), likely due to the change in ownership structure and land investments.

3 Data

Our newly built dataset combines information on all recorded episodes of expropriation with a comprehensive set of historical information on Italian municipalities, described below. Descriptive statistics for our preferred sample, discussed in Section 4.2, are reported in Table 1; additional details on the construction and sources of several variables are in Appendix Section C. Bianchi-Vimercati et al. (2025) includes all the data and code used in the analysis

[9] Already in 1960, an official inquiry of the Ministry for Agriculture found that 7% of the assignment contracts had been canceled, in roughly half the cases by the assignee and in a quarter mutually. Only one quarter of the cancellations were related to expulsions. The most common reasons for the cancellations, accounting for 64% of the total, were shifts to other more remunerative occupations and causes related to dislike of living in the country, apathy, and professional incapacity.

of this paper.

Expropriation

Our novel dataset includes each single land expropriation realized following the 1950 *Legge Stralcio*, extracting the first and last name of the expropriated landowner, municipality, and size of the expropriation from the original expropriation documents.[10] The official Italian government gazette published these documents between 1950 and 1953.

Our primary treatment variable is a binary indicator that equals one if a municipality was affected by the land reform (i.e., experienced any expropriation) and zero otherwise. To complement this, we construct an alternative continuous measure of treatment intensity, the percent expropriation, by aggregating the total expropriated land in each municipality and dividing it by the municipality's total area in 1951.[11] This measure is expressed in percentage points and allows us to examine heterogeneity in treatment effects across different levels of expropriation intensity. To the best of our knowledge, we are the first to digitize and analyze data on the intensity of land expropriation at the municipal level for the entire country.[12] Appendix Table A2 reports descriptive statistics for both treatment variables across all treated regions.

Land Distribution and Agricultural Income

We also digitized pre-reform data about the land distribution and agricultural income in Italy published in 1948 in Tables 1 and 2 of Medici (1948). Giuseppe Medici collected this data at the municipal level alongside the Italian Institute for Agricultural Economics. The effort was funded by the law DLL n. 381 of 1946, and data collection likely happened between April 1946 and 1948.

Socioeconomic and Political Variables

[10] For an example, see Appendix Figure C1.

[11] While dividing by cultivated land might better reflect the share of usable land, both historical accounts and empirical evidence suggest that land under cultivation is likely endogenous to economic and institutional conditions (see Section 2 and Martinelli, 2014).

[12] Albertus (2023) also digitized expropriation data at the plot level. Research for this paper and data digitization began independently of his and prior to that paper being available online. One important difference relates to the geographical scope: Albertus (2023) focuses on two regions, whereas we analyze all eleven regions with available data. Other substantial differences in scope, empirical strategy, and outcomes are discussed in Appendix Section F.

In our analysis, we rely on a broad set of socioeconomic indicators from decennial censuses, such as sectoral employment, resident population, and share of high school graduates. We use municipality-level data from the 1936-2001 Italian national censuses collected by the Italian Institute of National Statistics (ISTAT).

To measure long-term economic performance, we construct a municipality-level indicator of per capita income growth between 1970 and 2000. For the initial period, we digitized data on municipality-level total income estimates in 1970 from Bocca and Scott (1974), available through the Historical Archive of Banco di Roma. To the best of our knowledge, this is the earliest comprehensive snapshot of municipal-level economic conditions in post-war Italy. For the year 2000, we rely on official administrative data produced by the Ministry of Economics and Finance, which reports total income at the municipal level. We convert the 1970 measure to Euros in the year 2000 and calculate per-capita income by dividing both measures by the number of residents at the municipality level in the relevant year. We combine these two sources to compute a growth rate that captures changes in per capita income over a 30-year period. This approach allows us to assess the long-run economic impact of the land reform.

We construct a measure of land suitability for wheat cultivation at the municipal level using data from the Global Agro-Ecological Zoning (GAEZ) project developed by the Food and Agriculture Organization (FAO). Specifically, we calculate the average wheat suitability score for each municipality by spatially averaging grid-level GAEZ estimates across municipal boundaries. This measure reflects the agro-climatic potential for wheat production under rain-fed conditions and serves as a proxy for inherent land productivity prior to reform.

In addition, we measure land inequality using data from the decennial Italian Agrarian Censuses conducted by ISTAT in 1970, 1990, and 2000. These data provide detailed information on the distribution of landholdings across farms within municipalities. From these, we compute the Gini index of landholding size as a municipality-level indicator of land distribution inequality prior to and after the reform.

We complete our dataset with electoral data collected by the Ministry of Interior on the national elections from 1946 to 1987.

4 Empirical Strategy

4.1 Model

The panel structure of the data allows us to follow treated and untreated municipalities over time and estimate a difference-in-differences model under the assumption of parallel trends. The chosen model is:

$$y_{it} = \delta_i + \gamma_t + \alpha_{1936} \times d_{1936} \times E_i + \alpha_{Post} \times d_{Post} \times E_i + \varepsilon_{it} \tag{1}$$

where y_{it} is the economic outcome (e.g., share of agricultural employment) in municipality i in the year t; E_i represents either a treatment dummy or the percentage of expropriated lands; d_t are time dummies; δ_i and γ_t denote a full set of municipality and year-by-region fixed effects, respectively. This model controls for common changes over time within each region in the outcome through γ_t and for time-invariant, municipality-level characteristics through δ_i. We can then test for the presence of differential pre-trends (α_{1936}) and for the reform's impact after 1951 and until 2001 (α_{Post}). Alternatively, we can estimate dynamic effects over time by replacing $\alpha_{Post} \times d_{Post} \times E_i$ with $\sum_{\tau \in \{1936, \mathcal{T}^{post}\}} \alpha_\tau \times d_\tau \times E_i$ where $\mathcal{T}^{post}$ is the set of years after treatment. All coefficients are relative to 1951, whose coefficient is normalized to 0.[13] Our favorite specification uses clustered standard errors for each municipality to account for potential serial correlation at the municipal level, as is standard in difference-in-differences analysis.

[13] The reform law was approved on 28 July 1950 and pre-dates the 1951 census, set to represent a snapshot of the country as of November 4th, 1951. The implementation of the 1950 law, however, required some time: only 13 out of 1,143 of the digitized expropriation decrees related to the areas of interest were issued before the census, and the first was issued on August 30th, 1951. The actual reallocation likely took several months. We measure our main outcomes in 1936, 1951, 1961, 1971, 1981, 1991, and 2001.

4.2 Sample Definition

Our main analysis covers the entire country, but we also report separate estimates for Central-Northern and Southern Italy. This distinction is motivated by the contemporaneous implementation of the Cassa del Mezzogiorno (Casmez) program, which allocated substantial public transfers to southern municipalities to promote industrialization between 1952 and 1992 (see Colussi et al., 2021). These interventions could bias our estimates of the reform's effects. To address this threat to identification, we exclude municipalities that directly received Casmez funding during its implementation, and show sensitivity to their inclusion in the Appendix.

Showing separate estimates for the two areas has multiple advantages. The estimates for the South may still be affected by spillover effects and broader dynamics induced by the Casmez policy, and the two areas were characterized by different economic trends that predate the program. Finally, historical sources suggest that municipalities were strategically included or excluded from the reform in the South (see the archival evidence presented by Caprettini et al. (2024)). Although we address all these potential concerns via sample selection and year-by-region fixed effects, testing pre-trends separately for the two areas, among other robustness checks, allows the evaluation of the underlying parallel trend independently for each sample.

We estimate our model using two control group definitions. In our preferred sample, we compare all treated municipalities to non-treated ones in provinces with at least one treated unit. This control group is geographically proximate to the treatment group and is similarly affected by any region- and province-level policy. A map of the expropriated municipalities in treated provinces is in Figure 1. We consider this our preferred sample because it allows us to include all treated municipalities, thus providing greater statistical power and estimation precision.

Figure 2 shows how the land reform affected land inequality in treated areas for our preferred sample in the Center-North (top panels) and the South (bottom panels). Panels (a) and (c) show how land inequality changed in the treated and control areas after the land reform.

Panels (b) and (d) do the same for the share of land in large plots (over 50 hectares). We can observe partial convergence of both measures for the two groups of municipalities, consistent with a successful reform implementation. Appendix Table A3 reports the corresponding difference-in-differences estimates. We find that the reform significantly reduced the Gini index of land ownership and the share of land in large plots by meaningful amounts, in both the Center-North and South samples.

For the secondary control group definition, we employ a matching approach. The combination of difference-in-differences with matching has been introduced and discussed in Heckman et al. (1997), Heckman et al. (1998), Smith and Todd (2005), and Stuart et al. (2014). In the first step, we estimate the propensity score using a logit model based on the following baseline characteristics: land productivity in 1948, Gini index in 1948, number of agricultural firms in 1948, hectares of available agricultural land in 1948, number of manufacturing plants per 1,000 residents in 1951, resident population in 1951, soil suitability for wheat, municipal elevation, distance from the coast (linear and squared), longitude, and latitude.[14] Notably, pre-reform land productivity and the Gini index of land plot sizes are directly related to the extent of the expropriation (see Section 2 and Appendix Figure A1).[15] In the second step, we use a nearest-neighbor matching approach to find two similar non-treated municipalities for each treated one, based on the propensity score estimated in the first step. We also restrict matches to be in the same macro-region (South or Center-North). Twelve treated municipalities are not matched to control municipalities as their propensity scores are outside of the common support, and three still lack some of the matching variables. This leaves us with a sample consisting of 134 treated and 268 untreated municipalities.

This approach addresses concerns that treated municipalities are systematically different from those in the control group. For example, they tend to be closer to the coastline, a factor

[14] Productivity is measured as total agricultural income divided by hectares used for agriculture, both obtained from Medici (1948) for 1948. The same source provides plot size distributions, enabling the calculation of the Gini index.

[15] Medici (1948) lacks information on the agricultural income for some municipalities (42 out of 149 treated municipalities). To avoid a reduction in the available sample, we impute missing values based on the prediction from municipal elevation and soil suitability for the cultivation of wheat, tomatoes, rice, potatoes, oats, maize, and beans.

that might be relevant if proximity to ports influences the growth of the industrial sector. While the control group for our preferred sample is, on average, 15 kilometers farther from the coastline than the treatment group, matching produces a control group that is less than 3 kilometers farther, with a statistically insignificant difference. Additionally, this matching procedure mitigates concerns about the proximity of treatment and control municipalities, given that 46% of the municipalities in the matched control group are in provinces that didn't experience any expropriation. Table A4 in the Appendix shows the balance of the matched sample and of our preferred sample.

4.3 Identification

The identification of our model relies on the parallel trends assumption underlying the difference-in-differences approach (Angrist and Pischke, 2008). This assumption requires that, in the absence of the reform, the variables of interest would have evolved similarly across all municipalities. Figure 3 provides a visual representation of our event study estimates, indicating that the evolution of the key outcomes was highly similar between treatment and control areas before the reform in our preferred sample. This similarity likely stems from the control group selection criteria, which include only untreated municipalities located in provinces where at least one expropriation occurred. In all tables, we include the p-value of formal tests for differential pre-trends whenever pre-treatment observations are available. We find no statistically significant deviations from parallel trends prior to the reform for our main outcomes and most of the other outcomes in this paper, supporting the validity of our research design.

As discussed in Section 2, reform areas were chosen based on the recommendation of expert agronomists and the prevalence of large and inefficient land ownership. Consistent with expectations, Appendix Table A5 confirms that land inequality is the strongest predictor of expropriation, followed by average land productivity. In Section 5.1, we employ the doubly-robust approach proposed by Sant'Anna and Zhao (2020) and implemented in Callaway and Sant'Anna (2021). The results remain largely unchanged.

Finally, in Appendix D, we discuss how alternative identification strategies based on border discontinuities would suffer from low statistical power in this context. While the land reform was implemented within well-defined areas, municipalities near provincial borders experienced only limited expropriations. Thus, we could not identify a significant discontinuity in expropriations at the border.

5 Reform Effects on Sectoral Composition

Table 2 reports the estimated long-run effects of land reform on sectoral employment shares in the period 1936-2001, using both standard and matched samples. We present results separately for central-northern Italy, southern Italy, and the full national sample. For each sample, we estimate the post-reform effect of treatment on the share of individuals employed in agriculture (columns 1 and 4), manufacturing (columns 2 and 5), and other sectors (columns 3 and 6). Columns (1)-(3) report estimates from the full sample, while columns (4)-(6) present results using a matched sample based on pre-treatment characteristics.

We find strong positive effects on agricultural employment in the Center-North. Column (1) shows that areas subject to expropriation had 4.5 percentage points higher agricultural employment. Event study graphs in Figure 3 show that the effects emerge shortly after the reform and grow over time, indicating that while the agricultural sector contracted overall, treated areas retained a larger share of agricultural workers.[16] Column (2) assesses the impact on manufacturing employment, showing a negative effect of similar magnitude to that observed in agriculture. The largest decline occurs in 1981, followed by a gradual attenuation. These findings suggest that the land reform led to a relative contraction in manufacturing employment in treated areas. The effects on the tertiary sector are small and statistically insignificant across specifications. We observe a similar pattern in the matched sample (columns 4-6), though estimates are slightly smaller.

In contrast, estimates for the South are weaker and mostly statistically insignificant. While there is some suggestive evidence of an increase in agricultural employment that

[16]The event studies for the matched sample are reported in Appendix Figure A4.

was compensated by a decrease in the tertiary sector in the standard specification (column 3), most estimates are not statistically significant, especially in the matched sample. While lower statistical power due to a smaller sample could explain the lack of significance, spillovers from the Cassa del Mezzogiorno program are also consistent with these smaller, non-significant estimates for the southern regions.

The results for the full Italian sample largely mirror those of the Center-North sample, with statistically significant increases in agricultural employment and corresponding declines in manufacturing. The estimated effect on the tertiary sector remains modest and not statistically significant. Again, the matched sample yields consistent but slightly smaller effects.[17] The event study graphs in Figure 3 are similar to those for the Center-North sample: the impact starts appearing in the first period and peaks between 1981 and 1991.

Overall, our findings indicate that treated areas experienced a significant short-term increase in agricultural employment, which persisted over subsequent decades.[18] In Appendix Table A7 we show that the magnitude of this effect is proportional to the extent of expropriation, reinforcing the role of the land reform in shaping sectoral composition.[19] The observed increase in agricultural employment was offset by a corresponding decline in manufacturing employment, with the impacts growing over the first 30 years and persisting 50 years after the initial shock.

[17] Results remain consistent when we include municipalities affected by Casmez, as shown in Table A6. Note that a few municipalities in the Center-North, such as Isola d'Elba, also received Casmez funding, explaining the increase in the number of observations for the northern sample.

[18] We obtain consistent results when splitting the sample around the median of the share of agricultural workers in 1951. Results are available upon request.

[19] We performed a Sobel-Goodman-style mediation analysis on the specifications in columns (1) and (4) of Table 2, including the percentage of expropriated land along with the binary treatment. The share of the impact explained by expropriation varies depending on the sample (larger in the Center-North and smaller in the South) and the way we control for the expropriation percentage (uninteracted or interacted with region indicators). In the full sample (third panel, *Italy*), the share of impact explained by expropriation is between 47 and 53% in the uninteracted model and between 57 and 65% in the interacted model. In the North, where expropriation has a larger explanatory power, estimates range between 55 and 83% across models. Full results are available upon request.

5.1 Robustness

In this section, we perform robustness checks to address potential threats to our identification strategy. We test the sensitivity to alternative model specifications, samples, and inference assumptions.

First, we find that results are robust to the inclusion of additional time-varying controls. Specifically, we interact time effects linearly with a set of baseline municipal characteristics measured in 1951, including latitude, longitude, soil suitability for wheat, literacy rates, high school completion rates, and rurality. These adjustments account for differential trends based on geographic, educational, and agricultural conditions. The results, reported in Appendix Table A8, remain consistent in both magnitude and statistical significance.

We then show that our evidence is not driven by the inclusion in our sample of the administrative center of each province (see Appendix Table A9). Administrative centers are often the most populated town in the province and might have different economic dynamics. Estimated coefficients are virtually unchanged with respect to the baseline models.

Our treatment's geographical nature suggests that the expropriation intensity might be spatially correlated. Appendix Table A10 reports the baseline estimates with standard errors that account for spatial correlation using the procedure developed in Conley (1999)'s study. Specifically, columns (1) - (8) replicate odd columns of Table 2, with different distance cutoffs.[20] While standard errors are generally larger, overall significance patterns persist, especially for manufacturing.

To relax the assumption of unconditional parallel trends, we use the doubly robust estimator proposed by Sant'Anna and Zhao (2020) and condition on several predictors of expropriation, including land inequality and productivity (see Table A5). This estimator is consistent if the correct underlying model is a propensity score or an outcome regression model. Results for the average treatment effect on the treated are largely unaffected and very close to the main specification. Results are included in Appendix Table A11.

[20] We choose bandwidths of 5, 10 and 15 kilometers. The small dimension of Italian municipalities implies that over 30% of the municipalities in reformed provinces have their center within 5km of another municipality's center, and over 90% meet the same condition with a 15km bandwidth.

Finally, we provide an additional check addressing the differences in average distance to the coastline between treated and untreated municipalities in the treated province sample. To rule out that this might drive the main results, we show their robustness to excluding municipalities near the coast. Appendix Table A12 replicates our main specification for samples where a progressively larger number of municipalities is excluded. Point estimates remain within the initial confidence intervals, and significance is only marginally lowered, as expected following the decrease in the number of included municipalities. This, together with our findings from the matched sample, where we find substantial balance in the distance from the sea, supports our findings as not being driven by factors unrelated to the reform.

5.2 Mechanisms

In this section, we explore potential mechanisms underlying the persistent effects observed in both agriculture and manufacturing. We first explore whether, as suggested by prior literature, land redistribution favored an increase in human capital. Then we look at the effects of the reform on two agglomeration measures related to growth and industrialization: population density and geographical concentration. We then discuss mobility patterns induced by the reform and, finally, review the available evidence of the impact of changes in farming scale, agricultural productivity, and occupational mobility.

Education

Many studies have documented a positive relationship between land distribution and human capital development. Galor et al. (2009) develop a model in which economies with more equal land distribution implement public education earlier than economies characterized by more unequal distribution. Cinnirella and Hornung (2016) provides evidence of a negative relationship between landownership concentration and education in 19th-century Prussia. On the other hand, Albertus et al. (2020) show that a land reform implemented in the 1970s in Peru hindered human capital accumulation through "intergenerational rural stasis."

Using municipal-level data on educational outcomes, we examine the long-run effects of the Italian land reform on human capital. The results provide no consistent evidence of long-lasting impacts on education. This null result is notable given the widespread belief that early landownership structures often shape long-run human capital accumulation. Our findings suggest that the land reform and associated expropriations did not generate lasting effects on educational attainment across municipalities. This contributes to a growing recognition in the literature that null results can be valuable, particularly when they help reassess assumptions about historical persistence. As Abad and Maurer (2024) argue, the absence of long-term effects can be as revealing as their presence, especially when it invites a reconsideration of the mechanisms through which economic change translates into lasting development outcomes. Columns (1) and (5) of Table 3 report the estimated effect of the reform on the share of illiterate individuals. Across both the standard and matched samples, the coefficients are near zero and statistically insignificant, suggesting no measurable impact on basic literacy levels. Similarly, columns (2) and (6) examine the reform's effect on the share of individuals with completed high school. Again, the coefficients are statistically insignificant and vary in sign, indicating no clear relationship between expropriation and secondary education attainment. The only exception is in southern Italy, where we observe a positive and statistically significant effect. However, given the timing and location of other large-scale public interventions, especially the Cassa del Mezzogiorno, which financed major investments in infrastructure and industrial development, and their potential effect on returns to education in the nearby regions, this effect is unlikely to reflect the isolated impact of land reform. Instead, the estimated coefficient may partly capture the influence of broader state-led development policies that disproportionately benefited the South, potentially raising returns to education. This underscores the empirical difficulty of disentangling the effects of land reform from overlapping policy efforts in the southern regions. However, at the national level, the effect is not significantly different from zero, providing no systematic evidence that expropriation influenced educational attainment.[21]

[21] In columns (3) and (4) of Appendix Table A13, we use the identification approach described in Section 6 to study the reform's impacts on middle school completion. Middle school completion is not available in census data until 1981, preventing us from studying it with our preferred difference-in-differences approach. We find no impacts on the full sample and some negative impacts for the Southern sample. Columns (1), (2), (5), and (6) use the same approach for two outcomes included in Table 3 and find similar results. Overall, Appendix Table A13 shows that the reform was not associated with consistent changes in educational attainment.

Agglomeration

While usually jointly determined and modeled, agglomeration is generally considered to be necessary for industrialization and growth (see a variety of regional economics models discussed in Breinlich et al., 2014). In Table 3, columns (3), (4), (7), and (8) present the estimated effects of land reform on rurality and population density, using both standard and matched samples.[22] The results indicate that in treated municipalities, the reform was associated with a significant increase in rurality and a reduction in population density, particularly in the Center-North sample. These patterns suggest that the land reform weakened agglomeration forces, potentially contributing to the long-run shift away from manufacturing observed in our main results. While an initial increase in rurality is largely an expected consequence of the reform's settlement design, its persistence and the decline in population density capture broader demographic and spatial dynamics that are not mechanically driven by the housing structure. We take all the results as suggestive evidence that the Italian land reform may have disrupted agglomeration forces, slowing industrialization, which in turn further weakened local agglomeration dynamics.[23]

Migration

Redistribution has ambiguous effects on migration as it can provide capital necessary to migrate, tie beneficiaries to the land they received, and affect the availability of job opportunities in the area. It may also affect fertility and mortality. Historical sources don't provide strong support for meaningful changes in migration patterns. Angeli and INSOR (1979) notes that both treated and rural untreated areas experienced significant population declines between 1951 and 1971, and provides anecdotal evidence suggesting that the land reform did not fundamentally alter the migration patterns that characterized rural areas during this era.

In Table 4, we study how the shares of young (<15), working age (15-64), and older res-

[22]Population density is calculated as the ratio between the decennial population reported in the relevant census and the municipality's area in 1951, winsorized at 99%. Rurality is measured as the percentage of the total municipal population living in what the Italian Census classifies as *case sparse* (i.e., houses scattered across the municipality's territory without forming a residential nucleus).

[23]Agglomeration can generate very long term persistence, as shown in Bleakley and Lin, 2012.

idents (over 65) are affected by the reform. We find a one percentage point decrease in the share of younger people in the Center-North and the full sample. However, other changes in the age composition of the population are less robust and shrink dramatically or lose significance when switching to the matched difference-in-differences sample. We also find consistent reductions of around half a percentage point in the share of the male population. Since we consider youth younger than 15 to be unlikely to migrate in large numbers, we interpret these estimates as suggesting that reformed areas had reduced fertility, while migration (as suggested also by the reduction in density found in Table 3) disproportionally concerned men but had no clear impact on the age composition of the population.[24] The magnitude of these estimates appears too small for selective migration to explain a large part of the impact on sectoral employment in Table 2, especially because the share of the working-age population remained constant or even increased (see columns 2 and 6 in Table 4).

Due to data limitations, it is not possible to disentangle the contributions of fertility and migration to the demographic patterns we observe. The Italian Census reports population figures by broad age groups (under 15, 15-64, and 65+), but does not include information on individuals' municipality of birth. Moreover, municipality-level data on internal migration are not available for the period, making it infeasible to track mobility directly. As a result, we cannot assess whether the land reform influenced migration decisions differently for beneficiaries, potentially by increasing their ability to accumulate savings and migrate, as shown in Collins et al. (2024) and Galán (2024), or for non-beneficiaries, who may have been displaced by reduced local opportunities. Alternatively, the reform may have limited mobility by anchoring beneficiaries to newly acquired land, as discussed later in this section.

Farm Scale and Productivity

Another mechanism proposed in the literature relates to scale effects. Adamopoulos and Restuccia (2020) examine the effects of land reform in the Philippines, where, unlike in our context, a ceiling was imposed on landholdings, leading to a decline in agricultural produc-

[24] These findings are qualitatively and quantitatively comparable to those found by Caprettini et al. (2024) for a similar sample in the Center and North of Italy. However, our interpretations of the results differ as they suggest that the reduction of younger people was likely due to migration.

tivity. In Italy, while the reform reduced share of land in large plots (see Figure 2), no formal ceiling on ownership was ever implemented. Moreover, existing evidence indicates that productivity rose substantially in areas affected by the reform (see the discussion of King, 1973 in Section 2). In addition, Martinelli (2014) argue that, prior to the reform, large landowners in Italy operated as local monopsonists, deliberately restricting labor hiring relative to what would be observed under competitive labor market conditions. This suggests that land redistribution may have led to an increase in agricultural employment by shifting production to smaller farms that hired more labor.

While we cannot directly test the scale hypothesis in our setting, Figure 2 provides suggestive evidence that the reform reduced the preponderance of large plots, potentially affecting sectoral composition in line with prior literature.[25] Furthermore, occupational mobility, discussed later in this section, could introduce labor market frictions that may have increased reliance on family labor, consistent with theories linking scale and productivity (Foster and Rosenzweig, 2022).[26]

Another explanation for the observed divergence relates to the land improvement initiative carried out by the Italian Government alongside the land redistribution. Ginsborg (2003) highlights how the largest component of expenditures for land improvement (which represented 55% of the total, see Appendix Figure A3) was devoted to housing construction, while efforts to implement irrigation plans in some regions were largely unsuccessful. We also find that, while agricultural productivity was lower in reformed areas for the preferred sample, soil suitability was higher. Matching treated and control municipalities based on soil characteristics and pre-reform land productivity does not meaningfully affect estimates

[25] According to King (1973), the redistribution effort aimed to produce plot sizes that could support a family. This means that the average land plot size was larger in areas with lower potential productivity. Furthermore, heterogeneity analysis based on the share of land distributed as *poderi*, i.e., larger plots, does not yield conclusive results because those shares were very similar in most regions (between 72 and 86% except for Sardinia (55%). None of the estimates we obtained reached significance, likely due to a lack of statistical power. Results are available upon request.

[26] Foster and Rosenzweig (2022) propose and empirically test a model featuring a U-shaped relationship between productivity and plot size, which can reconcile our findings with those of Adamopoulos and Restuccia (2020). Their theory also suggests that smaller agricultural firms would employ relatively larger amounts of labor – often within the household – due to frictions in the labor market and economies of scale in agricultural machines.

(see columns 4-6 in Table 2).

Additional evidence of increased land productivity comes from the different uses of the land after the reform. According to Marciani (1966), in treated areas, arable land increased from 53.4% before the reform to 67.5% in 1963, while pasture and uncultivated land decreased from 35.4% to 16%. This change in distribution favored the expansion of cereal cultivation, paired with a significant increase in reared cattle quantity and quality, two higher-return land uses.

Occupational Mobility

Reduced occupational mobility can help explain the notable persistence of the reform's impact.[27] Angeli and INSOR (1979) surveyed farmers in reform areas and provides anecdotal evidence that the reform induced children of beneficiaries to work in agriculture[28], reflecting the scholarly work carried out in several other contexts.[29]

If children of agricultural workers are more likely to work in the same sector when their parents are landowners, land redistribution might have affected their occupational choices. Since a crucial aspect of the reform was to distribute land to agricultural workers, which would be inherited by their children, it potentially increased the probability that future generations would also work in agriculture. Long and Ferrie (2013) show that children of land-owning farmers have a probability 10 to 20 times higher of being farmers themselves relative to the children of fathers in other sectors (including landless agricultural workers) in the United States and Britain around the same time. To show that occupational mobility is affected by landownership in our context, we analyze data from the Italian Survey on Household Income and Wealth, which sampled Italian citizens in both treated and untreated

[27] Not all the above-discussed mechanisms have the potential to explain the observed persistence in agricultural specialization: for example, economies of scale are unlikely to deliver persistence independently, given that an optimal scale could be achieved or restored through land markets.

[28] Angeli and INSOR (1979) also argues that the land reform affected the age structure of landowners: according to a nationwide 1976 investigation, expropriated municipalities had more young male workers. Specifically, in municipalities without expropriations, men aged 14-49 working in agriculture were 37.9%, compared to 47.9% in municipalities with more than 20% of the entire territory expropriated.

[29] Fernando (2022) shows that Indian firstborns who inherit agricultural land display reduced migration and entry into non-agricultural sectors. Dunn and Holtz-Eakin (2000), among others, shows the transmission of self-employment. Corak and Piraino (2011) show that intergenerational transmission of employers is positively related to the presence of self-employment income. Lo Bello and Morchio (2021) highlights parental professional networks' role in occupational choices.

areas in the period 1977-2016. We find that male children of agricultural workers have approximately 40-50% higher probability of staying in agriculture when their parents own the land they are working on. Details for our analysis are available in Appendix E.

Anecdotal evidence, the extensive literature on occupational mobility, and our validation exercise suggest that the reform might have affected the occupational choices of its beneficiaries' children. This can explain a significant part of the reform's lasting impact on agricultural employment in the affected regions.

6 Reform Effects on Long-Run Economic Growth

Both reductions in scale and occupational mobility are related to inefficient outcomes. Foster and Rosenzweig (2022) estimate that if all Indian farms were at the minimum scale required to maximize the return on land, farm workers' income would rise by 68%. Caselli and Gennaioli (2013) argue that dynastic management, i.e., passing ownership and control of a firm from one generation to the other within a family, is a substantial driver of cross-country TFP differences. On the other hand, evidence of monopsony pre-reform (Martinelli, 2014), faster agricultural productivity growth in reformed areas (King, 1973), and the reduction of agency problems might lead to faster economic growth in reformed areas. Thus, the long-term effects of land reform on income growth remain an open empirical question.

To assess the impact of the reform on economic growth, we analyze municipal income per capita. Due to the lack of pre-reform income data, we employ a matching approach to estimate the reform's effects on per-capita income in 1970 and its growth between 1970 and 2000. Growth is calculated as the difference between income in 2000 and 1970 divided by income in 1970 (all income measures are expressed in Euros from the year 2000). We apply the same matching procedure described in Section 4.2 to identify comparable units for each reformed municipality based on their probability to be subject to the reform. We estimate this propensity score using a variety of features that historical sources relate to the reform, such as the Gini index of land ownership in 1948, calculated using Medici (1948), and a measure of productivity (the average pre-reform agrarian income from Medici, 1948).[30] The inclusion of all the key factors that determined expropriation intensity, like land distribution and productivity, in the propensity score used in the matching procedure is meant to fulfill the "back-door criterion" and provide identification.

Table 5 presents the estimated effects of land reform on income levels in 1970 and 2000, as well as on income growth between 1970 and 2000. Columns (1)-(2) show that the reform had no statistically significant effect on income levels in 1970, across both the Center-North and South. This suggests that any immediate economic benefits of land redistribution were probably limited. Columns (3)-(4) report the effects on income in 2000. The coefficients for the Center-North are negative and statistically significant at 10%. However, at the national level and in the South, the point estimates are not statistically significant.

The most striking result appears in columns (5)-(6), where we estimate the impact of land reform on long-run income growth. Across all samples and specifications, we find statistically significant and negative effects on long-run income growth, particularly in the Italy-wide sample, where treated municipalities experienced approximately 20 p.p. lower growth over 30 years. This represents a sizable reduction relative to the national average growth rate of approximately 100% over the same period. In the Center-North, the effect is around 14 p.p. and significant at the 5% level, while in the South, the negative effect is even larger (approximately 31-35 p.p.) and statistically significant at the 5-10% level, although the estimates are less precise due to the smaller sample size. The negative effects estimated in this matched sample are robust to controlling for pre-reform land productivity, land inequality (Gini index), and wheat suitability.

Our overall findings suggest that, through distortions in labor allocation and undermining the development of agglomeration, land redistribution not only affected the structural transformation process but also reduced long-run income growth.

[30]The full list of pre-reform covariates we use to estimate the propensity score is: land productivity in 1948, Gini index in 1948, number of agricultural firms in 1948, hectares of available agricultural land in 1948, number of manufacturing plants per 1,000 residents in 1951, resident population in 1951, soil suitability for wheat, municipal elevation, distance from the coast (linear and squared), longitude, and latitude. We also restrict matches to be in the same macro region (South or Center-North). We use a nearest neighbor procedure to identify two control municipalities for each treated one.

7 Conclusions

This study examines the long-term effects of Italy's post-WWII land reform, a large-scale redistribution effort with social, economic, and political objectives. Using administrative records, we construct a novel national dataset that captures exposure and intensity of redistribution at the municipal level, allowing for a detailed empirical analysis of its consequences.

First, we exploit this measure to evaluate the reform's impact on sectoral composition. Using a difference-in-differences framework, we find robust evidence that land redistribution led to a persistent increase in agricultural employment, accompanied by a decline in manufacturing employment. The persistent effects on structural change motivate an investigation of transmission mechanisms. The reform had limited effects on human capital accumulation, but contributed to a negative impact on agglomeration, with treated municipalities exhibiting persistently lower population density and more scattered housing than untreated ones. Historical sources, prior literature, and survey-based evidence suggest that the new ownership structure may have reduced occupational mobility and played a crucial role in shaping these outcomes.

Lastly, we assess the reform's impact on long-term economic development. We use a matching estimator and provide evidence of a negative relation between land reform exposure and income growth in the period 1970-2000. These findings suggest that, beyond the impact on wealth redistribution, the reform had limited short-term benefits in terms of economic development with large negative long-run effects on the local economies.

From a broader perspective, our results contribute to the ongoing debate on large-scale redistribution programs. We highlight how short-term redistributive policies may come at the cost of long-term economic distortions. In the Italian case, an initial reduction in inequality was followed by a slower structural transformation process and economic growth in subsequent decades. While generalizing from historical case studies requires caution, our findings suggest that land redistribution can exacerbate pre-existing labor market frictions and structural inefficiencies, ultimately hindering long-term economic growth.

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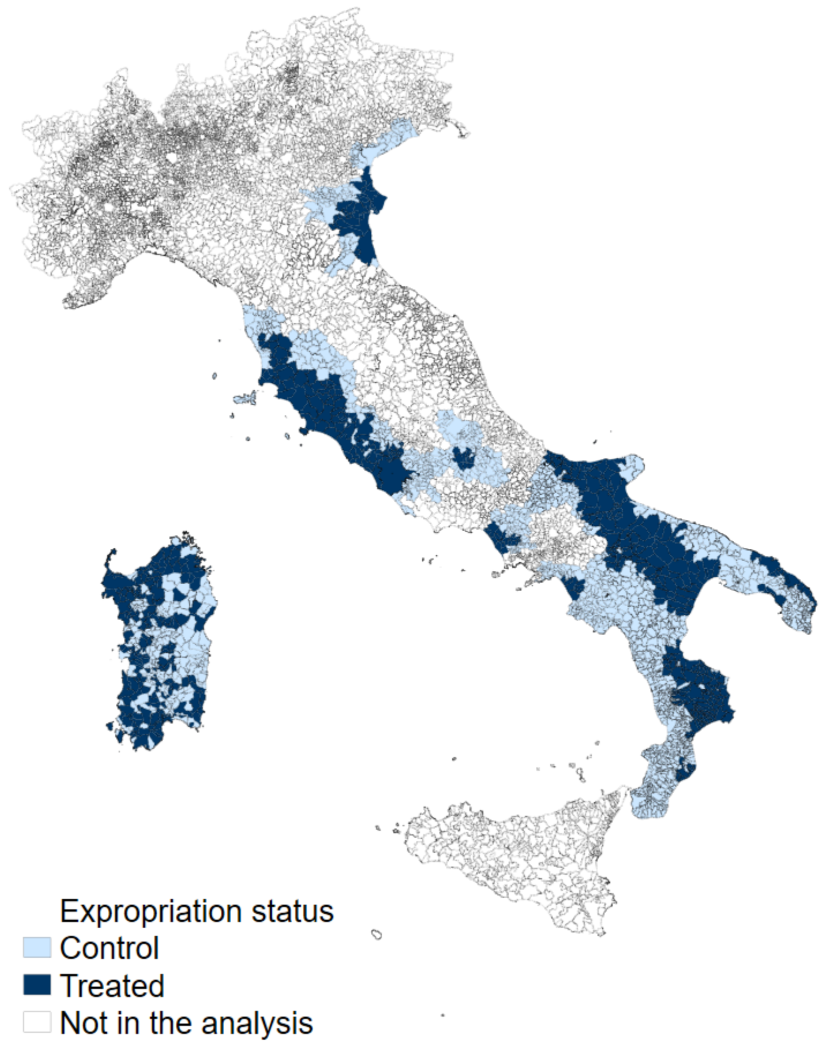
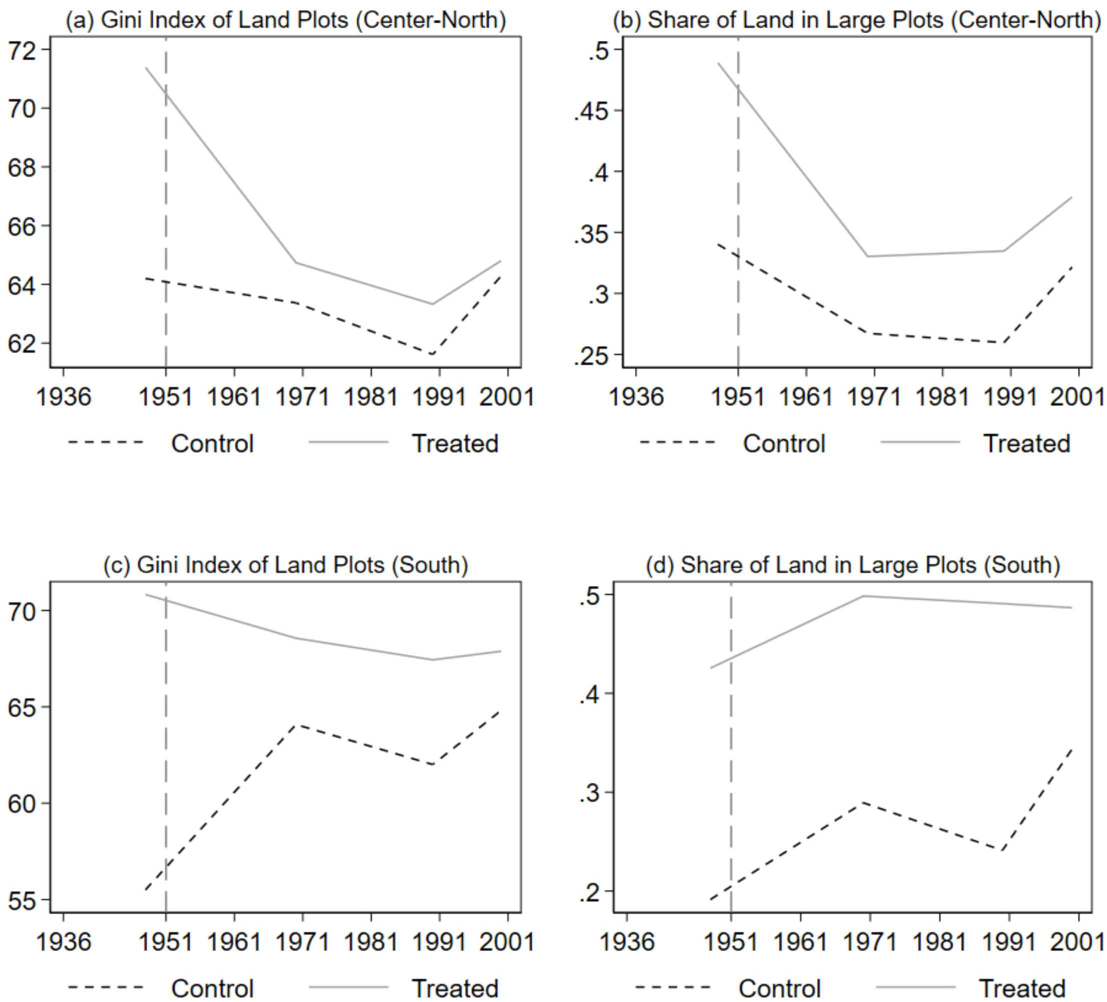
Paper Figures and Tables

Figure 1: Expropriated Municipalities in Treated Provinces



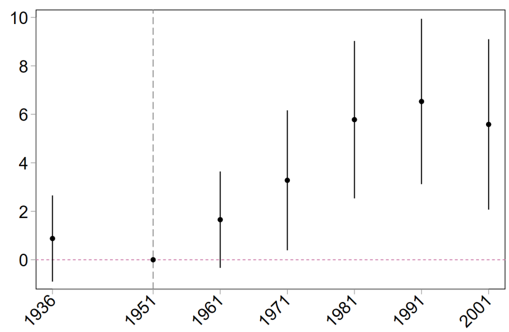
Notes. In dark blue, municipalities with at least one land reform expropriation; in light blue, municipalities in provinces where at least one municipality was expropriated. Light blue municipalities will comprise the main control group in our difference-in-differences analysis. *Source: Legge Stralcio.*

Figure 2: Land Inequality in Treated and Control Areas

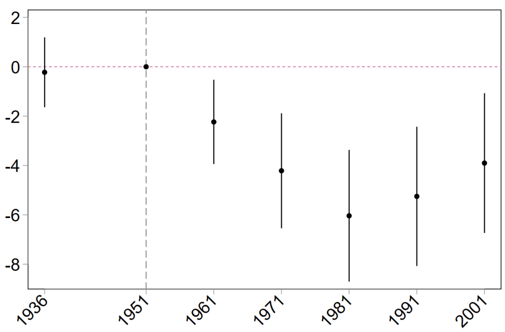


Notes. Panels (a) and (c) display the average Gini index of land plots in the Center-North and South samples, respectively, calculated using data from Medici (1948) and the Agricultural Censuses of 1970, 1990, and 2000. Panels (b) and (d) display the share of land belonging to plots larger than 50 hectares in the Center-North and South samples, respectively, as measured by the Population Census for 1936-2001. For all panels, we use the sample including all municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation for Italy.

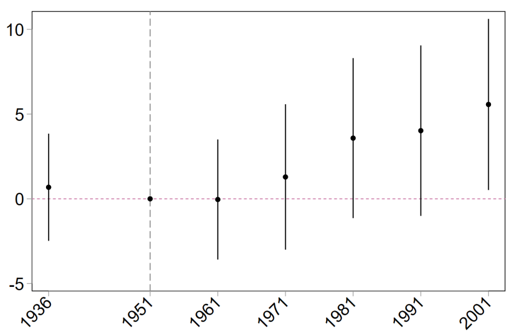
Figure 3: Difference-in-Differences Event Studies: Agriculture and Manufacturing



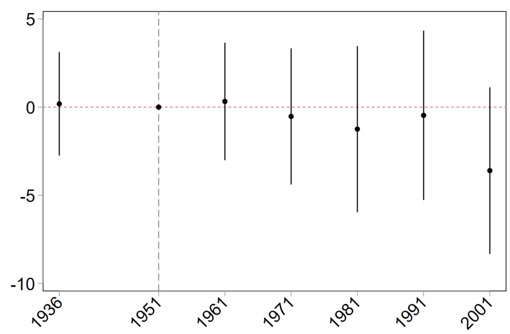
(a) Agriculture - Center-North



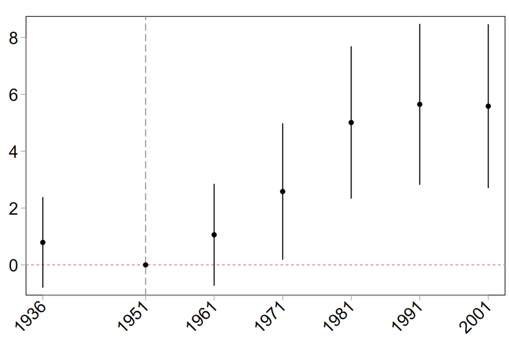
(b) Manufacturing - Center-North



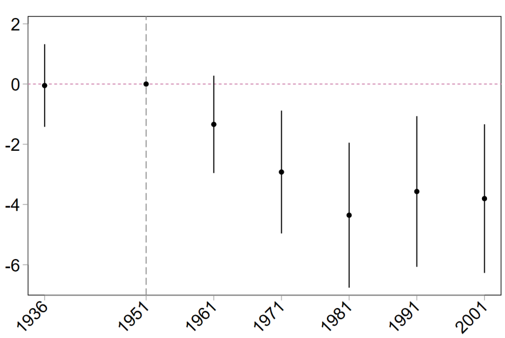
(c) Agriculture - South



(d) Manufacturing - South



(e) Agriculture - Italy



(f) Manufacturing - Italy

Notes. Event study estimates for the impact of land expropriation on agriculture and manufacturing employment. We use the standard sample, which includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation, and excludes municipalities that received funds from the Cassa del Mezzogiorno. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. Standard errors are clustered at the municipality level.

Table 1: Descriptive Statistics

	Mean	Min	Max	Std. Dev.	Observations
Expropriation					
Expropriation Dummy	0.22	0.00	1.00	0.41	2020
Land Expropriated (%)	0.02	0.00	0.99	0.07	2004
Census					
Empl. Agriculture (% - 1951)	68.07	2.96	97.94	17.89	1924
Empl. Manufacturing (% - 1951)	18.66	0.34	79.48	12.76	1924
Illiteracy Rate (1951)	0.20	0.02	0.58	0.08	2020
High School Completion Rate (1951)	1.79	0.10	12.47	1.11	1924
Female Population (% - 1951)	0.50	0.28	0.58	0.02	2020
Population Density (1951)	158.30	15.47	1486.62	203.55	1919
Political					
Christian-Democrat Votes (1948)	0.50	0.03	0.95	0.17	2020
Socialist-Communist Votes (1948)	0.25	0.00	0.89	0.20	2020
Geography					
Gini Index - Land Dist. (1948)	73.23	29.27	93.58	9.58	1820
Land Suitability (Wheat)	2116.88	24.00	8601.00	1709.06	2019
Municipality Elevation	347.67	0.00	1433.00	279.53	2020

Notes. We report summary statistics for the standard (non-matched) sample, which includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation. Municipalities that received funds from the Cassa del Mezzogiorno are excluded from the sample. Expropriation data and land distribution data digitized by the authors. Census data and electoral statistics are based on administrative data.

Table 2: Impact of Land Expropriation on Sectoral Employment

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform	4.566***	-4.328***	-0.239	3.562**	-3.538**	-0.0243
	(1.427)	(1.123)	(0.787)	(1.662)	(1.397)	(0.872)
Observations	2830	2830	2830	1815	1815	1815
Pretrend p-val.	0.335	0.752	0.171	0.708	0.723	0.158
South Sample						
Reform	2.884	-1.104	-1.780**	1.717	-0.635	-1.082
	(1.975)	(1.913)	(0.767)	(2.133)	(1.993)	(0.852)
Observations	2327	2327	2327	964	964	964
Pretrend p-val.	0.672	0.902	0.191	0.347	0.812	0.111
Italy						
Reform	3.977***	-3.198***	-0.778	2.852**	-2.420**	-0.432
	(1.159)	(0.997)	(0.580)	(1.310)	(1.153)	(0.629)
Observations	5157	5157	5157	2779	2779	2779
Pretrend p-val.	0.333	0.947	0.054	0.369	0.909	0.029
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using the standard sample, which includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation, and excludes municipalities that received funds from the Cassa del Mezzogiorno. Columns (4)-(6) estimate Model (1) using the matched sample resulting from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. We report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Impact of Land Expropriation – Human Capital and Agglomeration

	Standard				Matched			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% Illiterate	% High Sch.	Rurality	Pop. Density	% Illiterate	% High Sch.	Rurality	Pop. Density
Center-North Sample								
Reform	-0.0000314	-0.142	5.698***	-21.37***	-0.000835	-0.527	6.295***	-19.15*
	(0.00328)	(0.353)	(1.096)	(8.149)	(0.00372)	(0.345)	(1.354)	(9.759)
Observations	2490	2439	2439	2765	1566	1560	1560	1820
Pretrend p-val.	.	.	.	0.633	.	.	.	0.549
South Sample								
Reform	0.00191	0.899**	-1.199	0.216	0.00451	1.154**	-0.792	-5.173
	(0.00635)	(0.399)	(1.124)	(4.127)	(0.00776)	(0.464)	(1.269)	(4.228)
Observations	2130	2049	2049	2216	846	846	846	964
Pretrend p-val.	.	.	.	0.021	.	.	.	0.002
Italy								
Reform	0.000657	0.223	3.280***	-13.88**	0.00122	0.120	3.566***	-13.77**
	(0.00309)	(0.271)	(0.838)	(5.577)	(0.00375)	(0.283)	(0.988)	(6.258)
Observations	4620	4488	4488	4981	2412	2406	2406	2784
Pretrend p-val.	.	.	.	0.496	.	.	.	0.067
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(4) estimate Model (1) using the standard sample, which includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation, and excludes municipalities that received funds from the Cassa del Mezzogiorno. Columns (5)-(8) estimate Model (1) using the matched sample resulting from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936 (where available), 1951, 1961, 1971, 1981, 1991, and 2001 is included. Where available, we report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Impact of Land Expropriation – Population Composition and Migration

	Standard				Matched			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% <15	% 15-64	% >64	% Male	% <15	% 15-64	% >64	% Male
Center-North Sample								
Reform	-0.0122***	0.00955***	0.00261	-0.00363***	-0.0102***	0.00277	0.00745*	-0.00154
	(0.00260)	(0.00263)	(0.00341)	(0.00114)	(0.00293)	(0.00336)	(0.00409)	(0.00141)
Observations	2439	2439	2439	2439	1560	1560	1560	1560
South Sample								
Reform	0.000866	0.00998**	-0.0108***	-0.00638***	0.000214	0.00506	-0.00527	-0.00900***
	(0.00384)	(0.00427)	(0.00391)	(0.00237)	(0.00433)	(0.00477)	(0.00460)	(0.00270)
Observations	2049	2049	2049	2049	846	846	846	846
Italy								
Reform	-0.00759***	0.00970***	-0.00211	-0.00460***	-0.00621**	0.00365	0.00255	-0.00442***
	(0.00220)	(0.00227)	(0.00263)	(0.00112)	(0.00248)	(0.00276)	(0.00310)	(0.00137)
Observations	4488	4488	4488	4488	2406	2406	2406	2406
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(4) estimate Model (1) using the standard sample, which includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation, and excludes municipalities that received funds from the Cassa del Mezzogiorno. Columns (5)-(8) estimate Model (1) using the matched sample resulting from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936 (where available), 1951, 1961, 1971, 1981, 1991, and 2001 is included. Data to report the p-value for the estimate of placebo treatment in 1936 is unavailable. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Long-Run Income and Growth Impacts

	1970 Income		2000 Income		1970-2000 Growth	
	(1)	(2)	(3)	(4)	(5)	(6)
Center-North Sample						
Reform	30.98	112.5	-486.5*	-417.6*	-0.139**	-0.143**
	(133.1)	(124.2)	(212.3)	(195.4)	(0.0431)	(0.0450)
Observations	255	255	247	247	241	241
South Sample						
Reform	282.2	284.1	196.8	172.6	-0.350**	-0.316*
	(154.9)	(171.9)	(162.7)	(152.3)	(0.132)	(0.131)
Observations	137	137	99	99	95	95
Italy						
Reform	128.0	174.8	-287.0	-248.9	-0.200***	-0.210***
	(102.3)	(98.48)	(160.4)	(140.5)	(0.0501)	(0.0504)
Observations	392	392	346	346	336	336
Wheat Suit.	No	Yes	No	Yes	No	Yes
Gini Index	No	Yes	No	Yes	No	Yes
Productivity	No	Yes	No	Yes	No	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Estimates for the effect of the land reform on per capita income levels in 1970 and 2000, and on 1970-2000 growth. Income is always converted to 2000's Euros. Land suitability for wheat, Gini index, and land productivity are included as linear controls where indicated. We use the matched sample generated with the process discussed in Section 4.2. Standard errors are clustered following Conley (1999) with a 10km bandwidth.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Online Appendix for *Persistent Specialization and Growth: The Italian Land Reform*

by Riccardo Bianchi-Vimercati, Giampaolo Lecce and Matteo Magnaricotte.

A Additional Figures

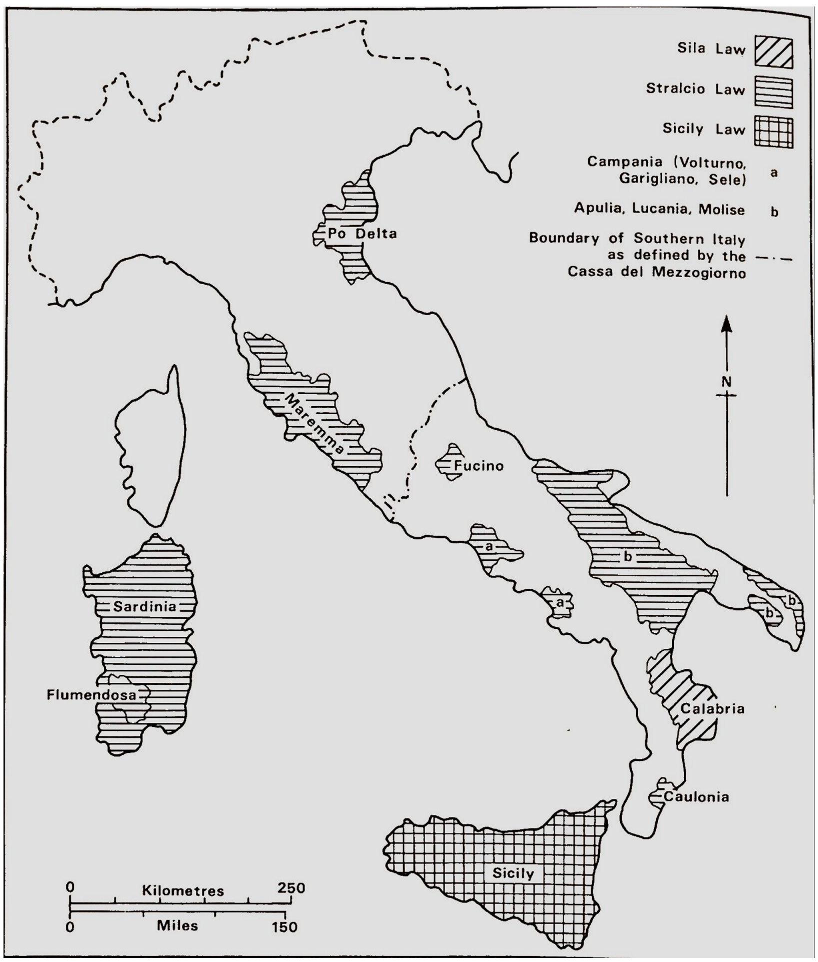
Figure A1: Expropriation Rule for the *Legge Stralcio*

Percentuali di scorporo riferite agli scaglioni di reddito imponibile

Scaglioni di reddito imponibile totale (Lire)	Imponibile medio per Ha. (Lire) 1000 e oltre	900	800	700	600	500	400	300	200	100 e meno
Fino a 30.000	—	—	—	—	—	—	—	—	—	—
Da oltre 30.000 a 60.000	—	—	—	—	—	0	15	30	55	70
» 60.000 a 100.000	—	—	—	—	0	10	30	60	70	85
» 100.000 a 200.000	35	40	47	55	60	65	70	75	84	90
» 200.000 a 300.000	45	50	55	60	65	70	75	80	87	95
» 300.000 a 400.000	52	57	60	65	70	75	80	85	90	95
» 400.000 a 500.000	60	64	66	71	76	80	85	90	95	95
» 500.000 a 600.000	64	70	76	78	80	85	90	95	95	95
» 600.000 a 700.000	68	74	79	82	85	90	95	95	95	95
» 700.000 a 800.000	72	78	82	85	90	95	95	95	95	95
» 800.000 a 900.000	76	82	86	90	93	95	95	95	95	95
» 900.000 a 1.000.000	82	86	90	93	95	95	95	95	95	95
» 1.000.000 a 1.200.000	90	92	95	95	95	95	95	95	95	95
Oltre 1.200.000	95	95	95	95	95	95	95	95	95	95

Notes. The percentage of land to be expropriated depended on a plot's total taxable income (vertical dimension) and average income per hectare (horizontal dimension). Landowners with higher income and lower productivity per hectare were expropriated larger shares of their land.

Figure A2: Areas interested by the land reform



Source: King (1973)

Figure A3: Breakdown of the expenses of the *Enti di riforma* in the 1950s

Tab. 41 - **Risultanze finanziarie per l'insieme degli Enti di riforma** (*a*) **a fine del decennio 1950-51 / 1959-60.**

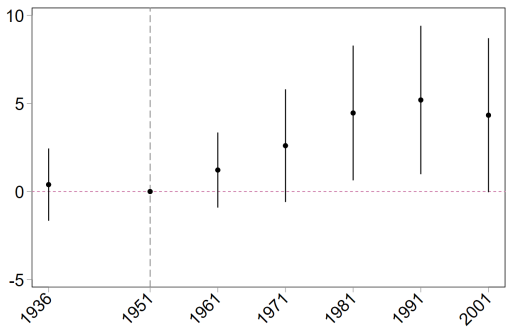
Categorie di entrata	Importi		Categorie di uscita	Importi	
	milioni di lire	%		milioni di lire	%
Assegnazioni per compiti istituzionali	512.760	*84,2*	Trasformazione fondiaria	340.006	*55,8*
			Assistenza e cooperazione	41.903	*6,9*
Redditi patrimoniali e entrate diverse	51.571	*8,5*	Acquisizione di macchine e scorte	38.661	*6,3*
Debiti verso banche	33.856	*5,5*	Anticipazioni per opere di bonifica eseguite in concessione	2.281	*0,4*
Totale entrate	**598.187**	***98,2***	Crediti verso assegnatari e cooperative	26.395	*4,3*
Disavanzo	10.867	*1,8*	Spese generali, di amministrazione e per oneri patrimoniali	138.101	*22,7*
			Interessi passivi	21.707	*3,6*
Totale a pareggio	**609.054**	***100,0***	**Totale uscite**	**609.054**	***100,0***

(*a*) Esclusa la Sezione speciale per la riforma fondiaria dell'Ente autonomo del Flumendosa.

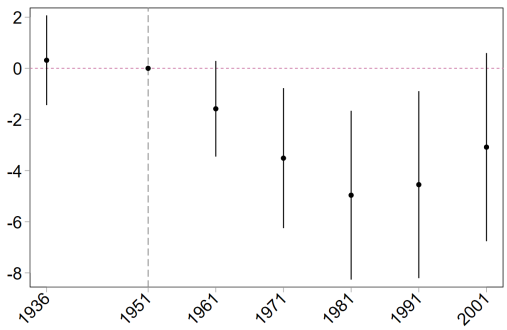
FONTE: Nostra elaborazione dei dati tratti dalle Relazioni della Corte dei conti al Parlamento (ATTI PARLAMENTARI, *citt.*).

Notes. Left column reports revenues; right column reports expenses. Among expenses, 55.8% is attributed to land transformation: historical reports (Ginsborg, 2003) report that most of the resources for land transformation were used to build new housing on the redistributed plots. General administrative costs of the reform accounted for 22.7% of the total. Source: Parliamentary Acts.

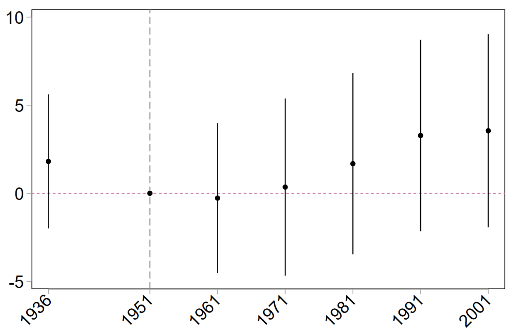
Figure A4: Matched Difference-in-Differences Event Studies: Agriculture and Manufacturing



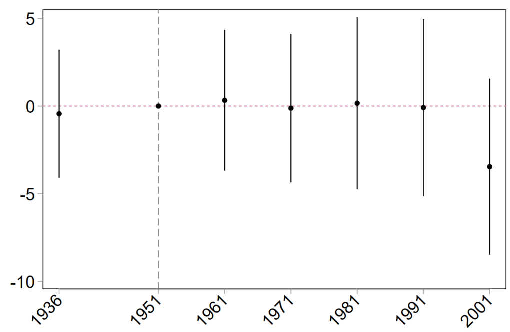
(a) Agriculture - Center-North



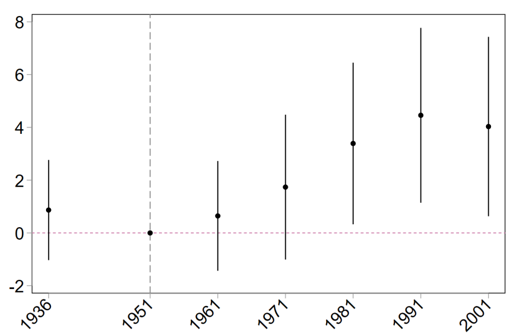
(b) Manufacturing - Center-North



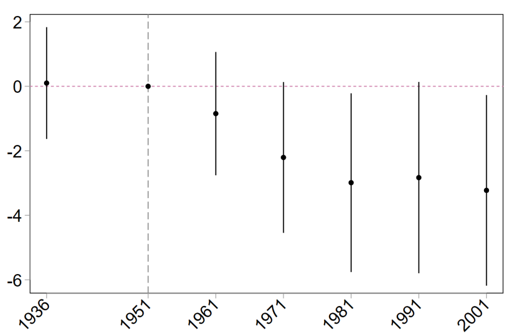
(c) Agriculture - South



(d) Manufacturing - South



(e) Agriculture - Italy



(f) Manufacturing - Italy

Notes. Event study estimates for the impact of land expropriation on agriculture and manufacturing employment. We use the matched sample, which results from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. Standard errors are clustered at the municipality level.

B Additional Tables

Table A1: Gross Saleable Production per Hectare

Year	Po Delta	Maremma	Fucino	Campania	A-L-M	Calabria	Sardinia	Total	Average (Italy)
1953	189	83	345	156	66	57	10	71	134
1954	182	81	275	133	55	60	15	73	129
1955	245	92	288	216	61	65	18	86	136
1956	226	97	292	242	63	80	20	90	133
I957	195	87	287	284	78	86	33	94	136
1958	247	110	379	280	89	98	48	114	151
1959	266	114	362	308	113	95	53	124	156
1960	246	107	375	330	92	98	56	116	151
1961	264	115	381	315	124	118	55	132	164
1962	265	135	414	411	138	129	59	148	165
1963	293	123	370	554	146	135	56	153	161
% yearly growth	**4**	**4.9**	**3.2**	**13.4**	**11.5**	**9.5**	**19.9**	**8.5**	**2.6**

Notes. Gross saleable production per ha. on assigned reformed lands (figures in ’000 lire, constant prices). *Source*: King (1973).

Table A2: Expropriation Data Statistic

Region	**Municipalities**	**Expropriations**	**Expropriated area (hectares)**	
			Total	Average
Center-North Area				
EMILIA-ROMAGNA	13 (44)	200	36,339.38	2,795.34
LAZIO	40 (180)	339	65,908.37	1,647.71
TOSCANA	38 (123)	533	115,823.7	3,047.99
VENETO	9 (93)	71	9,490.20	1,054.47
South Area				
ABRUZZO	9 (108)	17	15,874.06	1,763.79
BASILICATA	45 (131)	353	64,000.12	1,422.22
CALABRIA*	81 (262)	284	43,756.47	-
CAMPANIA	18 (262)	132	9,046.44	502.58
MOLISE	12 (84)	55	5,416.46	451.37
PUGLIA	60 (258)	1,108	129,062.08	2,151.05
SARDEGNA	113 (377)	240	45,554.93	403.14
Total	**438**	**3,332**	**540,272.93**	**-**

Notes. Values in parentheses report the overall number of municipalities in the treated provinces (i.e., provinces with at least one expropriation in their territory). Not all expropriation decrees in the Calabria region include the expropriated area: we report the total area expropriated in municipalities for which we have available data.

Table A3: Effect of Land Redistribution on Land Inequality

	Standard				Matched			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Gini Ind.	Gini Ind.	Large Plots	Large Plots	Gini Ind.	Gini Ind.	Large Plots	Large Plots
				Excluding Cassa				
Center-North Sample								
Reform	-6.596***	-35.20***	-0.0870***	-0.476***	-1.623	-18.52***	-0.0467*	-0.332***
	(1.268)	(5.287)	(0.0209)	(0.113)	(1.419)	(5.489)	(0.0241)	(0.118)
Observations	1210	1208	1210	1208	753	753	753	753
South Sample								
Reform	-7.004***	-43.78**	-0.0550	-0.684***	-0.185	10.20	0.0502	-0.154
	(2.329)	(18.75)	(0.0445)	(0.195)	(2.690)	(16.29)	(0.0532)	(0.218)
Observations	775	772	775	772	289	286	289	286
Italy								
Reform	-6.696***	-35.87***	-0.0792***	-0.493***	-1.225	-16.11***	-0.0199	-0.317***
	(1.115)	(5.101)	(0.0191)	(0.106)	(1.264)	(5.385)	(0.0228)	(0.110)
Observations	1985	1980	1985	1980	1042	1039	1042	1039
				Including Cassa				
Center-North Sample								
Reform	-6.613***	-35.41***	-0.0808***	-0.459***	-1.875	-19.15***	-0.0474**	-0.338***
	(1.234)	(5.207)	(0.0204)	(0.111)	(1.348)	(5.170)	(0.0224)	(0.112)
Observations	1286	1284	1286	1284	802	802	802	802
South Sample								
Reform	-7.076***	-34.45***	-0.109***	-0.716***	-3.233***	-10.44*	-0.0707***	-0.465***
	(0.771)	(7.478)	(0.0111)	(0.127)	(0.852)	(6.183)	(0.0124)	(0.0982)
Observations	4189	4153	4189	4153	2571	2545	2571	2545
Italy								
Reform	-6.962***	-34.81***	-0.102***	-0.619***	-2.921***	-13.61***	-0.0654***	-0.419***
	(0.656)	(5.082)	(0.00979)	(0.0825)	(0.726)	(4.561)	(0.0109)	(0.0708)
Observations	5475	5437	5475	5437	3373	3347	3373	3347
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Treatment	Dummy	%	Dummy	%	Dummy	%	Dummy	%

Notes. Columns (1), (2), (5), and (6) report estimates for the Gini index of land plots; Columns (3), (4), (7), and (8) report estimates for the share of land in large plots (over 50 hectares). The baseline period is 1948, and its data is obtained from Medici (1948); data for later years is obtained from the Italian Agriculture Censuses of 1990 and 2000. Year-by-region and municipality fixed effects are always included. Odd columns use a binary definition of treatment, while even columns define treatment as the percentage of land expropriated. Clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Treatment and Control Balance for Treated Province and for Matched Samples

Panel A: Treated Province Sample

	Control	Treatment	p-value
N	621	149	
Productivity	183.7	162.9	0.028
Gini of Land Ownership	74.4	82.3	<0.001
Distance from Coast (km)	27.7	12.8	<0.001
1951 Population	2475	4841	<0.001
Factories per 1000 residents	30.0	32.3	0.009
Wheat Soil Suitability	1405.1	1866.4	0.001

Panel B: Matched Sample

	Control	Treatment	p-value
N	268	134	
Productivity	174.1	162.3	0.18
Gini of Land Ownership	81.1	81.7	0.13
Distance from Coast (km)	20.3	17.3	0.12
1951 Population	3532.5	4229.5	0.099
Factories per 1000 residents	30.1	32.1	0.034
Wheat Soil Suitability	2463.3	1822.7	0.73

Notes. This table reports average values for several relevant dimensions in the two samples used for our difference-in-differences specifications. Averages are calculated separately for municipalities with at least one expropriation and municipalities unaffected by the reform. We test the difference between treatment and control along each dimension and report the resulting p-value. Details about the matching procedure are available in Section 4.2.

Table A5: Predicting Land Reform Intensity

	Political Predictors			Socioeconomic Predictors						Land Predictors			Geographical Predictors			All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Δ PCI Vote %	0.91***															0.62***
	(0.18)															(0.18)
Δ DC Vote %		0.03														-0.13
		(0.08)														(0.09)
Δ Turnout			0.32**													0.08
			(0.12)													(0.12)
Δ Agric. Empl. %				0.01												-0.38
				(0.13)												(0.40)
Δ Manuf. Empl. %					-0.10											-0.42
					(0.12)											(0.42)
Employment %						0.11										0.01
						(0.22)										(0.24)
Δ Density							-0.09									-0.11
							(0.15)									(0.14)
Literacy %								-0.14								-0.24
								(0.18)								(0.16)
High Educ. %									-0.50*							-0.41
									(0.22)							(0.23)
Gini Index										1.29***						1.09***
										(0.18)						(0.19)
Land Productivity											-1.45***					-1.12**
											(0.35)					(0.35)
Wheat Suitability												-0.66				-0.25
												(0.41)				(0.45)
Elevation													-0.30			-0.40
													(0.21)			(0.23)
Dist. from Coast														-0.59***		-0.41**
														(0.14)		(0.14)
Dist. from Rome															-1.14*	-2.09*
															(0.48)	(0.84)
N	767	767	767	767	767	767	767	767	767	767	767	767	767	767	767	767
Within R2	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.08	0.04	0.00	0.00	0.04	0.01	0.17

Notes. The outcome is the share of total land expropriated in the municipality. We use the standard sample, as in columns 1-3 of Table 2. All predictors are standardized. Whenever information is available for more than one pre-treatment observation, we use the change as a predictor (indicated by Δ). Predictors in columns 1-3 are from the Ministry of Interior for the national elections of 1946 and 1948; those in columns 4, 5, and 7 come from the national censuses of 1936 and 1951; those in columns 6, 8, and 9 come from the national census of 1951; those in columns 10, and 11 comes from Medici (1948); column 12 comes from FAO GAEZ; those in 13, 14, and 15 are obtained from geographical data. All columns include region fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Replication of Table 2 Including Municipalities receiving *Cassa del Mezzogiorno* Funds

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform	3.853***	-3.524***	-0.330	2.852*	-4.155***	1.303
	(1.479)	(1.142)	(0.805)	(1.689)	(1.457)	(0.842)
Observations	3005	3005	3005	1909	1909	1909
Pretrend p-val.	0.366	0.627	0.324	0.691	0.828	0.676
South Sample						
Reform	0.928	0.541	-1.469***	0.752	0.192	-0.945**
	(0.689)	(0.625)	(0.344)	(0.766)	(0.697)	(0.387)
Observations	10664	10664	10664	6664	6664	6664
Pretrend p-val.	0.430	0.422	0.866	0.749	0.853	0.747
Italy						
Reform	1.591**	-0.381	-1.210***	1.194*	-0.722	-0.472
	(0.631)	(0.552)	(0.323)	(0.702)	(0.632)	(0.353)
Observations	13669	13669	13669	8573	8573	8573
Pretrend p-val.	0.723	0.543	0.707	0.664	0.834	0.607
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using a sample that includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation. Unlike the main sample, this one includes municipalities that received funds from the Cassa del Mezzogiorno. Columns (4)-(6) estimate Model (1) using the matched sample resulting from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. We report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Replication of Table 2, Using the Share of Land Expropriated as Treatment

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Share Expropriated	34.34***	-30.30***	-4.036	23.89*	-20.09*	-3.800
	(12.69)	(9.593)	(7.069)	(14.31)	(11.31)	(7.779)
Observations	2825	2825	2825	1815	1815	1815
Pretrend p-val.	0.316	0.657	0.253	0.685	0.852	0.268
South Sample						
Share Expropriated	38.94	-21.68	-17.26	17.70	-3.095	-14.60
	(27.10)	(30.14)	(10.76)	(26.78)	(26.61)	(12.19)
Observations	2320	2320	2320	957	957	957
Pretrend p-val.	0.857	0.389	0.223	0.739	0.812	0.317
Italy						
Share Expropriated	35.14***	-28.80***	-6.338	22.76*	-17.00	-5.766
	(11.50)	(9.502)	(6.124)	(12.72)	(10.51)	(6.753)
Observations	5145	5145	5145	2772	2772	2772
Pretrend p-val.	0.457	0.853	0.104	0.618	0.744	0.130
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using a sample that includes municipalities affected by the reform and all the unaffected municipalities in provinces with at least one expropriation. We exclude municipalities that received funds from the Cassa del Mezzogiorno. Columns (4)-(6) estimate Model (1) using the matched sample resulting from the procedure presented in Section 4.2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. We report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Replication of Table 2 with Additional Linear Time Controls

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform	4.157***	-4.101***	-0.0560	4.008**	-3.478***	-0.530
	(1.242)	(0.899)	(0.787)	(1.589)	(1.156)	(1.008)
Observations	2751	2751	2751	1813	1813	1813
Pretrend p-val.	0.088	0.343	0.178	0.359	0.925	0.184
South Sample						
Reform	2.970*	-0.998	-1.973**	2.479	-0.905	-1.574*
	(1.709)	(1.643)	(0.776)	(1.927)	(1.854)	(0.866)
Observations	2222	2222	2222	964	964	964
Pretrend p-val.	0.591	0.953	0.185	0.166	0.537	0.085
Italy						
Reform	3.710***	-2.978***	-0.732	2.943**	-2.130**	-0.813
	(0.975)	(0.834)	(0.577)	(1.147)	(0.988)	(0.652)
Observations	4973	4973	4973	2777	2777	2777
Pretrend p-val.	0.368	0.804	0.030	0.402	0.728	0.018
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes
Other Time Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using the standard sample, while columns (4)-(6) estimate it using the matched sample as in Table 2. Year-by-region and municipality fixed effects are always included. We also control for time effects interacted linearly with latitude, longitude, soil suitability for wheat, literacy rates, high school completion rates, and rurality, all measured in 1951. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. We report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Robustness Excluding Administrative Centers of Each Province

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform	4.912***	-4.556***	-0.356	3.886**	-3.834***	-0.0521
	(1.428)	(1.116)	(0.804)	(1.655)	(1.383)	(0.884)
Observations	2767	2767	2767	1787	1787	1787
Pretrend p-val.	0.398	0.775	0.217	0.727	0.689	0.147
South Sample						
Reform	2.884	-1.104	-1.780**	1.717	-0.635	-1.082
	(1.975)	(1.913)	(0.767)	(2.133)	(1.993)	(0.852)
Observations	2327	2327	2327	964	964	964
Pretrend p-val.	0.672	0.902	0.191	0.347	0.812	0.111
Italy						
Reform	4.188***	-3.324***	-0.864	3.046**	-2.594**	-0.451
	(1.160)	(0.997)	(0.587)	(1.308)	(1.148)	(0.634)
Observations	5094	5094	5094	2751	2751	2751
Pretrend p-val.	0.382	0.968	0.069	0.383	0.880	0.028
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using the standard sample, while columns (4)-(6) estimate it using the matched sample as in Table 2. In all samples, we remove the administrative centers of the province (*capoluogo di provincia*). Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. We report the p-value for the estimate of placebo treatment in 1936 to test the presence of potential pre-trends. In parentheses, we report standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Robustness: Conley Standard Errors

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform (5km)	4.566***	-4.328***	-0.239	3.562**	-3.538**	-0.0243
	(1.541)	(1.188)	(0.916)	(1.625)	(1.439)	(0.871)
Reform (10km)	4.566**	-4.328***	-0.239	3.562*	-3.538**	-0.0243
	(1.993)	(1.447)	(1.243)	(1.940)	(1.693)	(1.098)
Reform (15km)	4.566**	-4.328***	-0.239	3.562*	-3.538*	-0.0243
	(2.232)	(1.579)	(1.447)	(2.082)	(1.868)	(1.222)
South Sample						
Reform (5km)	2.884	-1.104	-1.780**	1.717	-0.635	-1.082
	(2.133)	(2.062)	(0.792)	(2.041)	(1.898)	(0.845)
Reform (10km)	2.884	-1.104	-1.780**	1.717	-0.635	-1.082
	(2.214)	(2.217)	(0.727)	(2.131)	(2.062)	(0.755)
Reform (15km)	2.884	-1.104	-1.780***	1.717	-0.635	-1.082
	(2.582)	(2.621)	(0.682)	(2.500)	(2.505)	(0.732)
Italy						
Reform (5km)	3.977***	-3.198***	-0.778	2.852**	-2.420**	-0.432
	(1.250)	(1.065)	(0.661)	(1.273)	(1.154)	(0.628)
Reform (10km)	3.977***	-3.198***	-0.778	2.852**	-2.420*	-0.432
	(1.506)	(1.225)	(0.856)	(1.445)	(1.316)	(0.743)
Reform (15km)	3.977**	-3.198**	-0.778	2.852*	-2.420	-0.432
	(1.696)	(1.366)	(0.984)	(1.589)	(1.493)	(0.812)
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using the standard sample, while columns (4)-(6) estimate it using the matched sample as in Table 2. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included. Standard errors obtained using the procedure in Conley (1999) for different bandwidths are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Double-Robust Estimation Replicating Columns (1) and (2) of Table 2 and Controlling for Expropriation Predictors for Full Sample

	(1)	(2)	(3)	(4)	(5)
ATT Agr. %	4.685***	4.026***	3.828***	3.951**	3.568**
	(1.299)	(1.346)	(1.381)	(1.647)	(1.676)
ATT Man. %	-3.899***	-3.443***	-2.336**	-4.179***	-2.314**
	(1.120)	(1.141)	(1.099)	(1.183)	(1.163)
Observations	4979	4979	4979	4979	4979
Political	No	Yes	No	No	Yes
Geography	No	No	Yes	No	Yes
Land	No	No	No	Yes	Yes

Notes. Estimates of Model 1 using the doubly-robust difference-in-differences estimator described in Sant'Anna and Zhao (2020) and implemented in the `did` package by Callaway and Sant'Anna (2021). The sample used is for all municipalities in provinces with at least one expropriation, and excludes municipalities that received funds from the Cassa del Mezzogiorno. Reported coefficients are the average of all treatment effects estimated post-reform. Different columns control for different combinations of the significant reform predictors from column 16 in Table A5. Political predictors include the change in PCI vote shares, Geography predictors include distance from the coast and from Rome, Land predictors include Gini index and land productivity. Clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Robustness: Excluding Coastal Municipalities

	Standard			Matched		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agr. %	Man. %	Other %	Agr. %	Man. %	Other %
Center-North Sample						
Reform (1m)	3.658**	-4.218***	0.560	3.624*	-4.869***	1.245
	(1.437)	(1.185)	(0.796)	(1.841)	(1.559)	(0.917)
Reform (10km)	3.620**	-4.838***	1.217	3.586*	-5.897***	2.311**
	(1.666)	(1.319)	(0.931)	(2.047)	(1.680)	(1.090)
Reform (15km)	3.501**	-4.677***	1.176	3.203	-5.733***	2.531**
	(1.770)	(1.383)	(1.013)	(2.133)	(1.722)	(1.192)
South Sample						
Reform (1m)	-0.605	2.228	-1.623**	-0.815	2.699	-1.884**
	(1.822)	(1.824)	(0.788)	(2.210)	(2.204)	(0.932)
Reform (10km)	-2.211	4.224**	-2.013**	-3.049	3.969*	-0.920
	(1.907)	(1.849)	(0.887)	(2.480)	(2.333)	(1.216)
Reform (15km)	-0.834	2.362	-1.527*	-0.0755	0.467	-0.391
	(2.199)	(1.969)	(0.920)	(2.799)	(2.401)	(1.193)
Italy						
Reform (1m)	2.105*	-1.870*	-0.235	1.840	-1.828	-0.0126
	(1.140)	(1.032)	(0.588)	(1.421)	(1.313)	(0.673)
Reform (10km)	1.551	-1.623	0.0713	1.298	-2.495*	1.197
	(1.290)	(1.136)	(0.694)	(1.615)	(1.421)	(0.837)
Reform (15km)	2.159	-2.499**	0.339	2.276	-3.981***	1.705*
	(1.409)	(1.168)	(0.767)	(1.727)	(1.430)	(0.929)
Mun. FE	Yes	Yes	Yes	Yes	Yes	Yes
YearXRegion FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Columns (1)-(3) estimate Model (1) using the standard sample, while columns (4)-(6) estimate it using the matched sample as in Table 2. We progressively exclude municipalities based on their distance from the coast. Year-by-region and municipality fixed effects are always included. Data for the census years 1936, 1951, 1961, 1971, 1981, 1991, and 2001 is included.

Table A13: Impacts on Education in 2001

	% Illiterate		% Middle Sch.		% High Sch.	
	(1)	(2)	(3)	(4)	(5)	(6)
Center-North Sample						
Reform	0.003*	0.002*	0.610	0.399	-1.787	-1.352
	(0.001)	(0.001)	(0.585)	(0.523)	(1.112)	(0.956)
Observations	261	261	260	260	260	260
South Sample						
Reform	-0.003	-0.003	-2.692**	-2.741**	2.574***	2.816***
	(0.004)	(0.004)	(0.902)	(1.024)	(0.772)	(0.830)
Observations	141	141	141	141	141	141
Italy						
Reform	0.000	0.000	-0.661	-0.773	-0.108	0.101
	(0.002)	(0.002)	(0.525)	(0.495)	(0.768)	(0.681)
Observations	402	402	401	401	401	401
Wheat Suit.	No	Yes	No	Yes	No	Yes
Gini Index	No	Yes	No	Yes	No	Yes
Productivity	No	Yes	No	Yes	No	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Estimates for the effect of the land reform on education outcomes measured in 2001 (share illiterate, share with middle school diploma or higher, share with high school diploma or higher). Land suitability for wheat, Gini index, and land productivity are included as linear controls where indicated. We use the matched sample generated with the process discussed in Section 4.2. Standard errors are clustered following Conley (1999) with a 10km bandwidth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

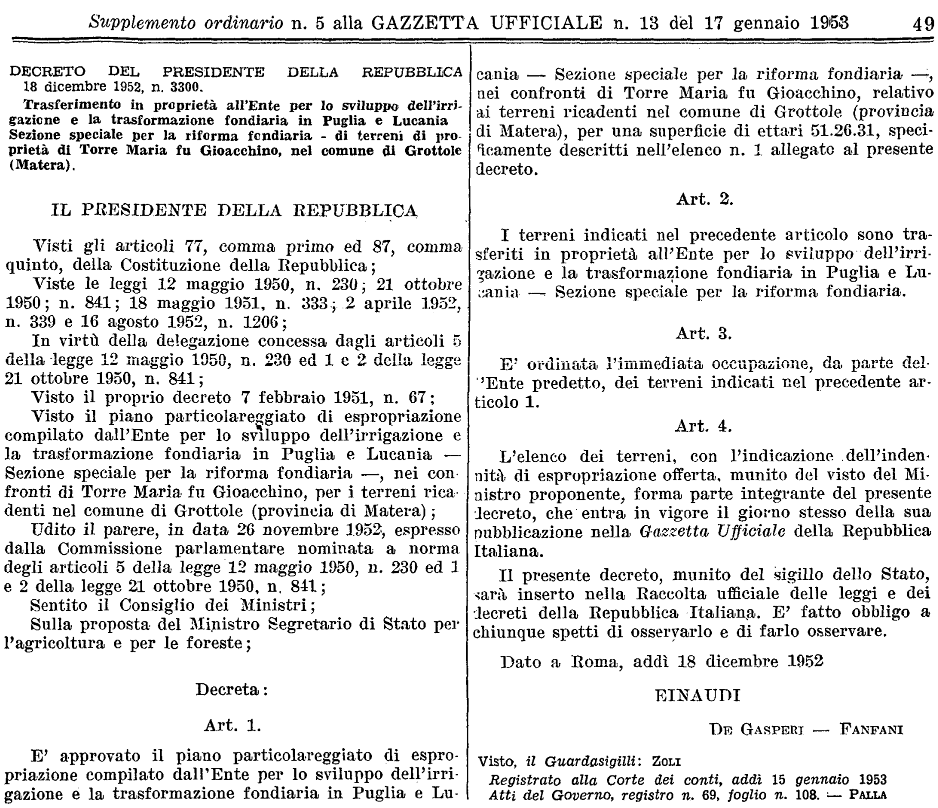
C Data: Description and Sources

In this section, we report the definition and source of the main variables used in the empirical analysis. Unless otherwise stated, the authors were responsible for the digitization of analog sources.

Expropriation Data

Treatment variables have been digitized from original expropriation documents (i.e., *Gazzetta Ufficiale*). For each expropriation, we collected information on the first name and the last name of the beneficiary, municipality, and size of the expropriation. Figure C1 reports an example.

Figure C1: Example of reported expropriation in *Gazzetta Ufficiale*



Supplemento ordinario n. 5 alla GAZZETTA UFFICIALE n. 13 del 17 gennaio 1953 49

DECRETO DEL PRESIDENTE DELLA REPUBBLICA 18 dicembre 1952, n. 3300.

Trasferimento in proprietà all'Ente per lo sviluppo dell'irrigazione e la trasformazione fondiaria in Puglia e Lucania Sezione speciale per la riforma fondiaria - di terreni di proprietà di Torre Maria fu Gioacchino, nel comune di Grottole (Matera).

IL PRESIDENTE DELLA REPUBBLICA

Visti gli articoli 77, comma primo ed 87, comma quinto, della Costituzione della Repubblica;

Viste le leggi 12 maggio 1950, n. 230; 21 ottobre 1950; n. 841; 18 maggio 1951, n. 333; 2 aprile 1952, n. 339 e 16 agosto 1952, n. 1206;

In virtù della delegazione concessa dagli articoli 5 della legge 12 maggio 1950, n. 230 ed 1 e 2 della legge 21 ottobre 1950, n. 841;

Visto il proprio decreto 7 febbraio 1951, n. 67;

Visto il piano particolareggiato di espropriazione compilato dall'Ente per lo sviluppo dell'irrigazione e la trasformazione fondiaria in Puglia e Lucania — Sezione speciale per la riforma fondiaria —, nei confronti di Torre Maria fu Gioacchino, per i terreni ricadenti nel comune di Grottole (provincia di Matera);

Udito il parere, in data 26 novembre 1952, espresso dalla Commissione parlamentare nominata a norma degli articoli 5 della legge 12 maggio 1950, n. 230 ed 1 e 2 della legge 21 ottobre 1950, n. 841;

Sentito il Consiglio dei Ministri;

Sulla proposta del Ministro Segretario di Stato per l'agricoltura e per le foreste;

Decreta:

Art. 1.

E' approvato il piano particolareggiato di espropriazione compilato dall'Ente per lo sviluppo dell'irrigazione e la trasformazione fondiaria in Puglia e Lucania — Sezione speciale per la riforma fondiaria —, nei confronti di Torre Maria fu Gioacchino, relativo ai terreni ricadenti nel comune di Grottole (provincia di Matera), per una superficie di ettari 51.26.31, specificamente descritti nell'elenco n. 1 allegato al presente decreto.

Art. 2.

I terreni indicati nel precedente articolo sono trasferiti in proprietà all'Ente per lo sviluppo dell'irrigazione e la trasformazione fondiaria in Puglia e Lucania — Sezione speciale per la riforma fondiaria.

Art. 3.

E' ordinata l'immediata occupazione, da parte dell'Ente predetto, dei terreni indicati nel precedente articolo 1.

Art. 4.

L'elenco dei terreni, con l'indicazione dell'indennità di espropriazione offerta, munito del visto del Ministro proponente, forma parte integrante del presente decreto, che entra in vigore il giorno stesso della sua pubblicazione nella *Gazzetta Ufficiale* della Repubblica Italiana.

Il presente decreto, munito del sigillo dello Stato, sarà inserto nella Raccolta ufficiale delle leggi e dei decreti della Repubblica Italiana. E' fatto obbligo a chiunque spetti di osservarlo e di farlo osservare.

Dato a Roma, addì 18 dicembre 1952

EINAUDI

DE GASPERI — FANFANI

Visto, *il Guardasigilli*: ZOLI

Registrato alla Corte dei conti, addì 15 gennaio 1953
Atti del Governo, registro n. 69, foglio n. 108. — PALLA

We define a municipality as treated if it experienced at least one expropriation. We also calculate the total area expropriated and divide it by the municipality area in 1951 to calculate an alternative treatment definition.

Data on Sectoral Employment

Employment in Agriculture % is the percentage of people over 15 years old in the agricultural sector over total workforce. *Source:* Data for 1936 are digitized from the Population Census (ISTAT) while data for the period 1951-2001 are from the 8000census database, available at https://ottomilacensus.istat.it/.

Employment in Manufacturing % is the percentage of people over 15 years old in the manufacturing sector over total workforce. *Source:* Data for 1936 are digitized from the Population Census (ISTAT) while data for the period 1951-2001 are from the 8000census database, available at https://ottomilacensus.istat.it/.

Employment in Other % is the percentage of people over 15 years old that are not classified as agricultural and manufacturing sector over total workforce (i.e. mainly service sector). *Source:* Data for 1936 are digitized from the Population Census (ISTAT) while data for the period 1951-2001 are from the 8000census database, available at https://ottomilacensus.istat.it/.

Income Data

Income 1970 is an estimation of the average income at the municipal level in 1970, expressed in 2000 euros. The data are from Bocca and Scott (1974). The primary source is the Historical Archive of Banco di Roma, specifically *Banco di Santo Spirito*. We digitized information on total income and total number of residents to obtain per capita income at the municipal level.

Income 2000 is an estimation of the average income at the municipal level in 2000, expressed in 2000 euros. The data were downloaded from the Ministry of Economy and Finance. We calculate per capita income as the ratio between the overall taxable (IRPEF) income over the population in each municipality from the 2001 Census.

Other Variables

Population Density is computed as the ratio between the population reported in a specific population census and the municipality's area in 1951. Our measure is winsorized at 99%. *Source:* Data for 1936 are digitized from the Population Census (ISTAT) while data for the period 1951-2001 are from the 8000census database, available at https://ottomilacensus.istat.it/.

Rurality is the percentage of the population living in *nucleo abitato* (i.e., a tiny nucleus of houses in the territory of the municipality) or in *case sparse* (i.e., houses spread over the territory of the municipality but without forming a residential nucleus) over the total population at municipal level. *Source:* Data are from the 8000census database, available at https://ottomilacensus.istat.it/.

Share of People with Completed High School is the share of people in the population (aged 6 and above) that completed at least high school. *Source:* Data are from the 8000census database, available at https://ottomilacensus.istat.it/.

Illiteracy Rate is the share of people in the population (aged 6 and above) that is illiterate. *Source:* Data are from the 8000census database, available at https://ottomilacensus.istat.it/.

Population Composition by age is divided in three distinct groups: the share of young (i.e. younger than 15), working age (15-64), and older residents (i.e. over 65) *Source:* Data are from the 8000census database, available at https://ottomilacensus.istat.it/.

Land Gini Index is computed using the land distribution reported for each municipality. *Source:* Raw data on the land distribution for 1948 are digitized from Medici (1948). Raw data on the land distribution for 1970 is digitized from the 1970 Census of Agriculture. Raw data on the land distribution for 1990 and 2000 is from the digital databases of the respective Censuses of Agriculture.

Land Productivity in 1948 is computed using the land distribution and the agricultural income reported for each municipality. *Source:* Raw data are digitized from Medici (1948).

Wheat Suitability is computed using the the average wheat suitability score for each municipality by spatially averaging grid-level GAEZ estimates across municipal boundaries. *Source:* Raw data are from Global Agro-Ecological Zones (GAEZ) project run by the FAO (FAO-GAEZ dataset, version 4.0). https://gaez.fao.org/.

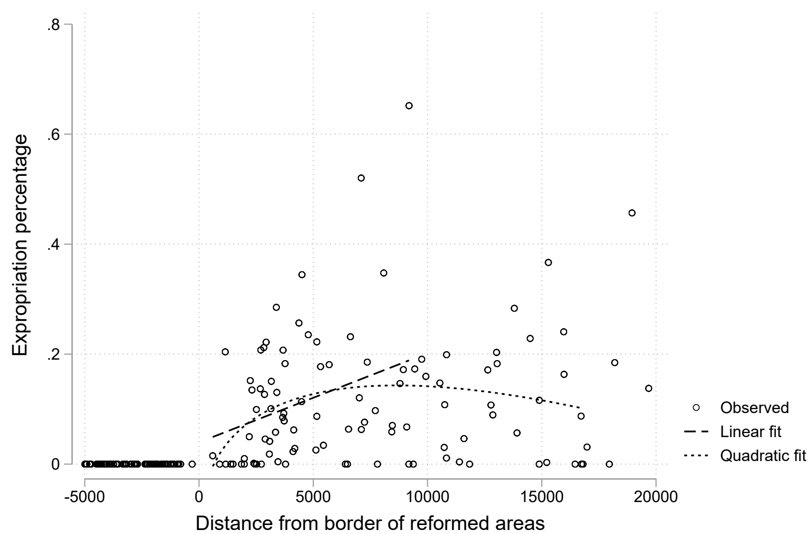
Vote shares for major parties and turnout for elections between 1946 and 1987 are obtained as shares out of total votes and out of the eligible population, respectively. *Source:* Raw data are from the Italian Ministry of Interior.

Cassa del Mezzogiorno We received a list of municipalities that have received transfers from the Cassa del Mezzogiorno between 1950 and 1992. *Source:* We thank the authors of Colussi et al. (2021) for kindly sharing their data.

D Additional Results: Regression Discontinuity

In this section, we provide evidence against the use of a regression discontinuity design for our empirical strategy. Our main treatment variable is the expropriation of land at the municipality level. Looking at the spatial distribution of expropriated land can inform us on the magnitude of the discontinuity at the border of municipalities that were part of reformed areas. Figure D1 shows the scatter plot of the expropriation data, ranked by distance from the closest border of the reformed areas. The figure also displays a linear and quadratic fit within the bandwidths chosen by the procedure described in Calonico et al. (2017), shown in Table D1. A visual inspection of the plot reveals the absence of a sharp discontinuity at the border for our main treatment variable. Therefore, employing a regression discontinuity design based on the distance from reformed areas would not capture the underlying variation we want to capture well.

Figure D1: Expropriation percentage and distance from the border of reform areas



Notes. The y-axis represents the percentage of land in the municipality expropriated by the reform; the x-axis reports distance from the reform border, where negative values mean municipalities were not treated. The positive slope of the linear and quadratic fits of the data and the small discontinuity around 0 suggest that a Regression Discontinuity Design would be statistically underpowered to identify the effects of land redistribution.

In Table D1, we formally test the discontinuity in the percentage of expropriated land using distance from the border of reformed areas as our running variable. We do so for a linear and a quadratic specification, which correspond to the two fitted lines displayed in Figure D1. In line with state-of-the-art techniques on regression discontinuity designs (Cattaneo et al., 2019), the Table reports the conventional estimate of the local treatment effect at the discontinuity, with the corresponding optimal choice for the bandwidth. The Table also shows conventional and robust standard errors, where the latter accounts for bias. The expropriation percentage does not display a significant discontinuity at the threshold.

Table D1: Regression Discontinuity

	Expropriation %	
	Linear	Quadratic
Distance	0.029	0.012
Conventional s.e.	(0.026)	(0.035)
Robust s.e.	(0.033)	(0.041)
Bandwidth (m)	9330	16787
Observations	1581	1581

Notes. Treatment is a binary variable taking value of 1 for municipalities within the reform borders. Outcome is the percentage of land expropriated by the reform. The two columns control for linear or quadratic effects of distance from the reform border. Estimates of the change in outcome at the discontinuity are not significant at standard thresholds when using the bias-robust standard errors implemented by the `rdrobust` package by Calonico et al. (2014). This suggests that a Regression Discontinuity Design would be statistically underpowered to identify the effects of land redistribution.

E Occupational Mobility Analysis

To show that occupational inheritance is affected by landownership in our context, we analyze data from the Italian Survey on Household Income and Wealth, which sampled Italian citizens in both treated and untreated areas in the period 1977-2016. This survey not only allows us to identify the occupational sector of young adults and their parents, but also reports whether each person owns the land they are working.

The outcome of interest in our analysis is a binary indicator for employment in agriculture. The main independent variable is a binary indicator of whether an individual's father ever worked in agriculture as a business owner; we build and include the same variable for other sectors. Finally, we control for the sector where the father was last employed, the year of the survey, and the respondent's age. We include males aged 20 or older and estimate the following linear probability model:

$$agr_{it} = \beta_1 agr\_owner_i + \beta_2 other\_owner_i + \sum_s \theta_s \{father\_sector_i = s\} + \theta_t + \rho age_i + e_i \quad (2)$$

Columns 1 and 2 of Table E1 show that ownership of land is positively related to occupational transmission in the agricultural sector for men. Male children of agricultural workers have approximately 40-50% higher probability of staying in agriculture when their parents own the land they are working on. On the contrary, we find that parental business ownership in other sectors is linked to a lower probability of employment in agriculture. Results are very similar when extending the analysis to include female respondents (see columns 3 and 4).

Table E1: Occupational Mobility Analysis

	Men only		Men and Women	
	(1)	(2)	(3)	(4)
Business Owner (Agriculture)	0.0548***	0.0442***	0.0403***	0.0306***
	(0.0150)	(0.0145)	(0.0110)	(0.0108)
Business Owner (Other Sector)	-0.0199	-0.0195	-0.0162*	-0.0165*
	(0.0134)	(0.0126)	(0.00981)	(0.00947)
Region FE	No	Yes	No	Yes
Mean (Father in Agriculture)	0.104	0.104	0.0769	0.0769
Mean (Father in Other Sector)	0.0675	0.0675	0.0556	0.0556
Observations	3040	3040	4902	4902

Notes. Data from the Bank of Italy's Survey on Household Income and Wealth (SHIW). Our sample consists of males older than 19 surveyed 1977-2016 in columns (1)-(2) and both males and female in columns (3)-(4). All columns include controls for age, survey year, and father's last employment sector (6 categories). Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

F Studies of the Italian Land Reform

We outline below how our approach differs in terms of data, identification strategy, and empirical focus from two papers studying other aspects of the Italian land reform, Percoco (2018) and Albertus (2023).

Albertus (2023) uses a geographic discontinuity approach and focuses on two regions, Tuscany and Lazio, to offer evidence of negative long-term effects on well-being as measured in 2011. Percoco (2018) evaluates the impact of the land reform in three regions in the South of Italy. Using a Kitagawa-Oaxaca-Blinder approach, it finds improved development in reformed areas. Our paper builds on both studies and addresses some of their limitations.

First, we collected and digitized the universe of plot-level expropriation data: rather than comparing areas which *could* be targeted by the reform, we are able to focus on those that experienced actual land redistribution within a difference-in-differences framework and relate its intensity to our outcomes of interest.[31]

Another important difference comes from the empirical design. Our difference-in-differences approach controls for municipality-level fixed effects, which enables us to effectively account for systematic differences between treated and untreated municipalities; we also address potential spatial spillovers by using different control group definitions that include municipalities geographically distant from the treated areas.

An important difference comes from the geographical scope. We offer a substantially more comprehensive evaluation of the reform by analyzing all eleven regions (four in the Center-North, and seven in the South) for which expropriation data are available. The only excluded region where land redistribution occurred is Sicily, due to the lack of expropriation data.

Including the South of Italy in the analysis requires attention to a major policy that happened at the same time and potentially confounds the land reform's impacts. Unlike previous studies, we use data on transfers from the *Cassa del Mezzogiorno*, a concurrent indus-

[31] Albertus (2023) also digitized expropriation data at the plot level. Research for this paper began independently of his and prior to that paper being available online. The two studies have substantial differences in scope, empirical approach, and outcomes studied, as discussed in this Section.

trialization policy aimed at the Southern regions, to tackle this identification challenge. We present results excluding affected municipalities and find that the coefficients for manufacturing employment change in a meaningful way (compare Table 2 and Appendix Table A6), although the difference is not significant.

The additional data, the increased geographical scope, and the robust approach produce novel findings and a nuanced picture. Our estimates show that the reform consistently increased the share of agricultural employment while decreasing the share in manufacturing, with stronger impacts in the Center and North of Italy; in the South, we find suggestive evidence for a larger reduction in tertiary sector employment. We track these outcomes over 50 years, showing the evolution over time of the reform's impact. We also produce previously unavailable evidence on the mechanisms driving these findings, suggesting that a reduction in agglomeration constrained the structural transformation of these local economies, while overall human capital accumulation was not significantly affected. We test the robustness and sensitivity of our results to a large variety of changes in samples and specifications and discuss any remaining potential limitations.

Finally, newly digitized income estimates at the municipality level allow us to produce the first estimates of the reform's effects on local economic growth in the medium- and long-term. We find that the reform led to 20 percentage points lower income growth between 1970 and 2000, with negative effects in both the Center-North and South samples.